

BFS Traversal

Step 1: Start at A

Legend

Complete

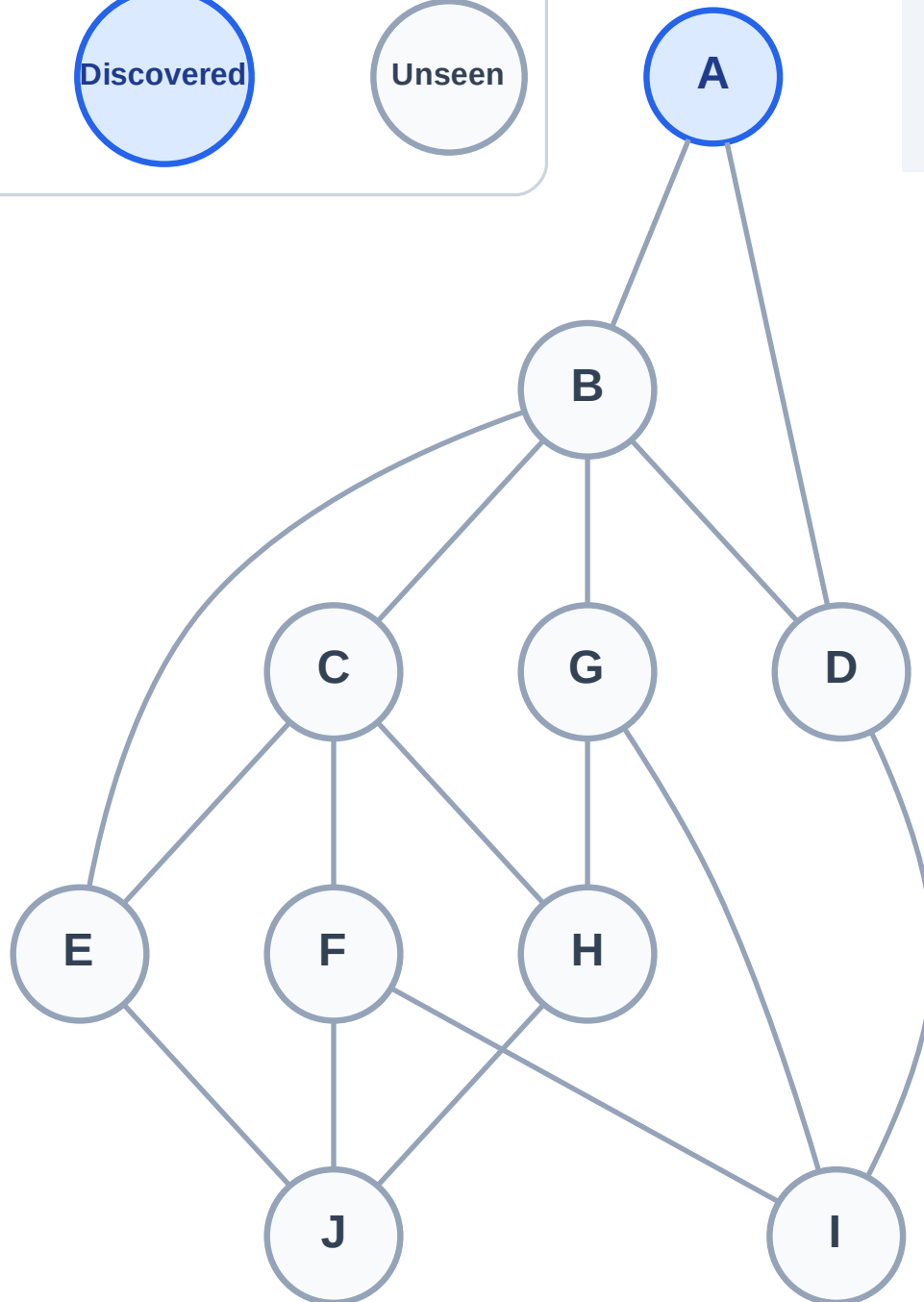
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

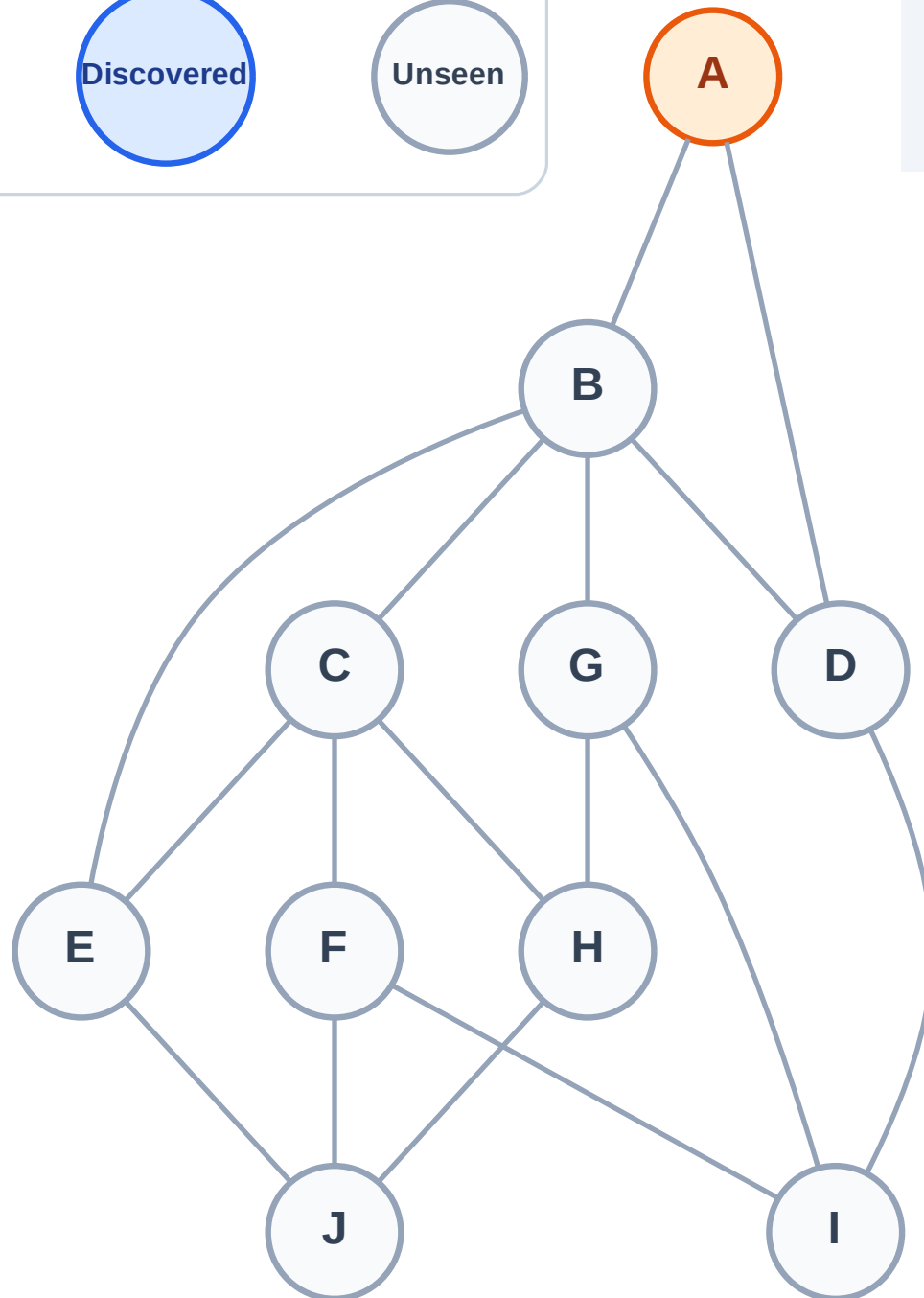
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

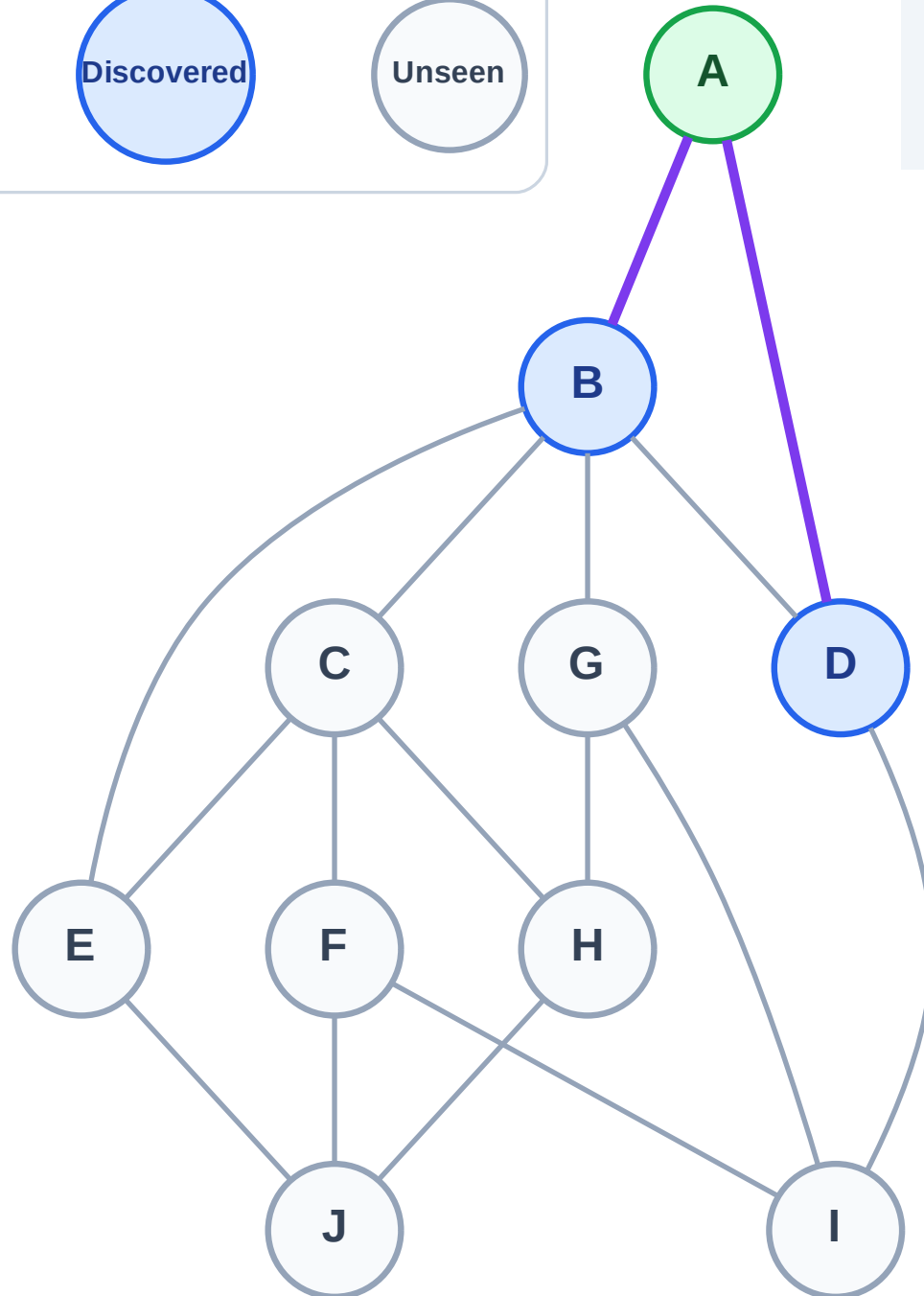
Step 3: Discover B, D from A

Legend



Queue: B → D

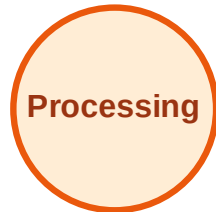
Visited: A



BFS Traversal

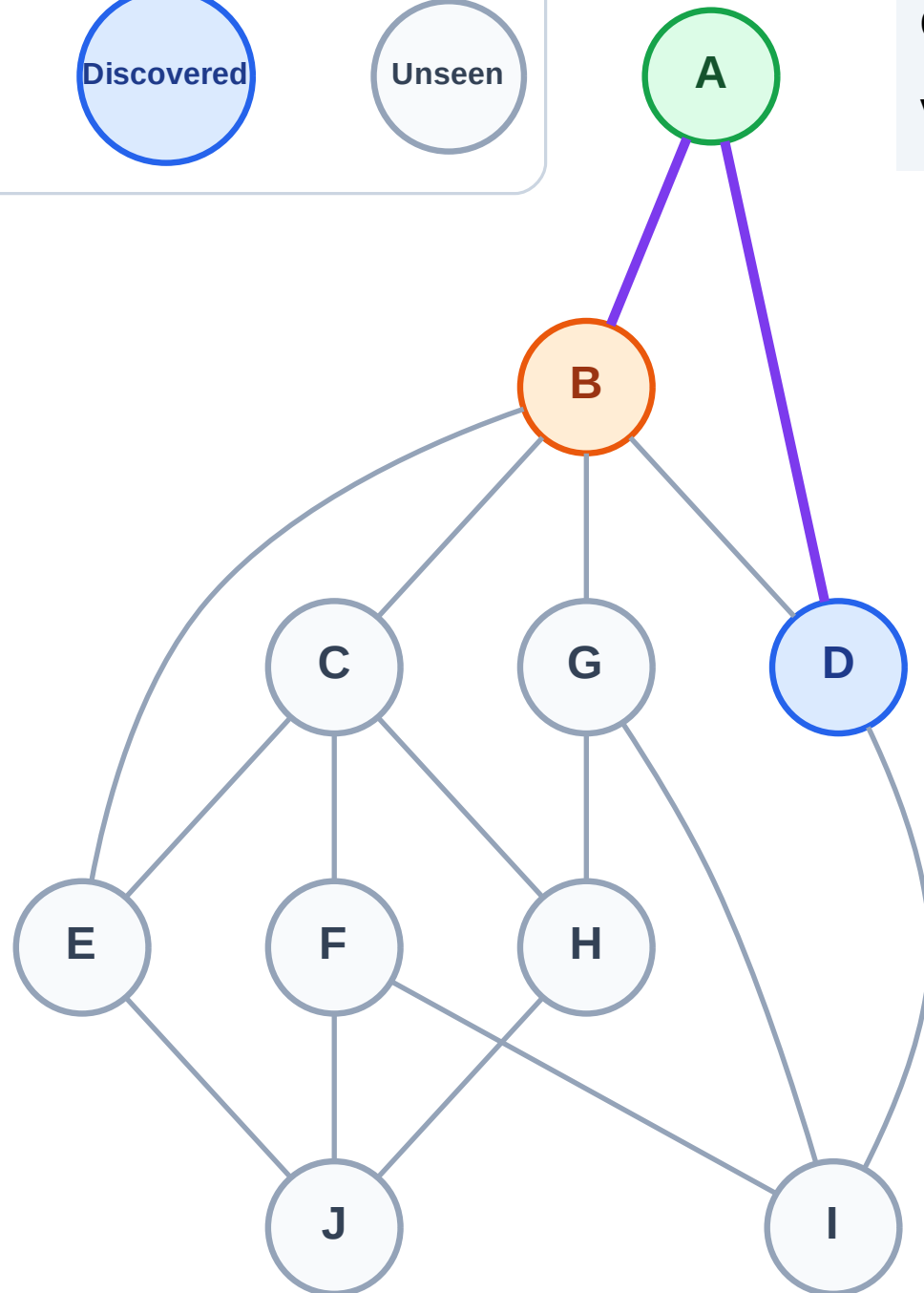
Step 4: Process node B

Legend



Queue: D

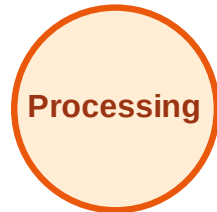
Visited: A



BFS Traversal

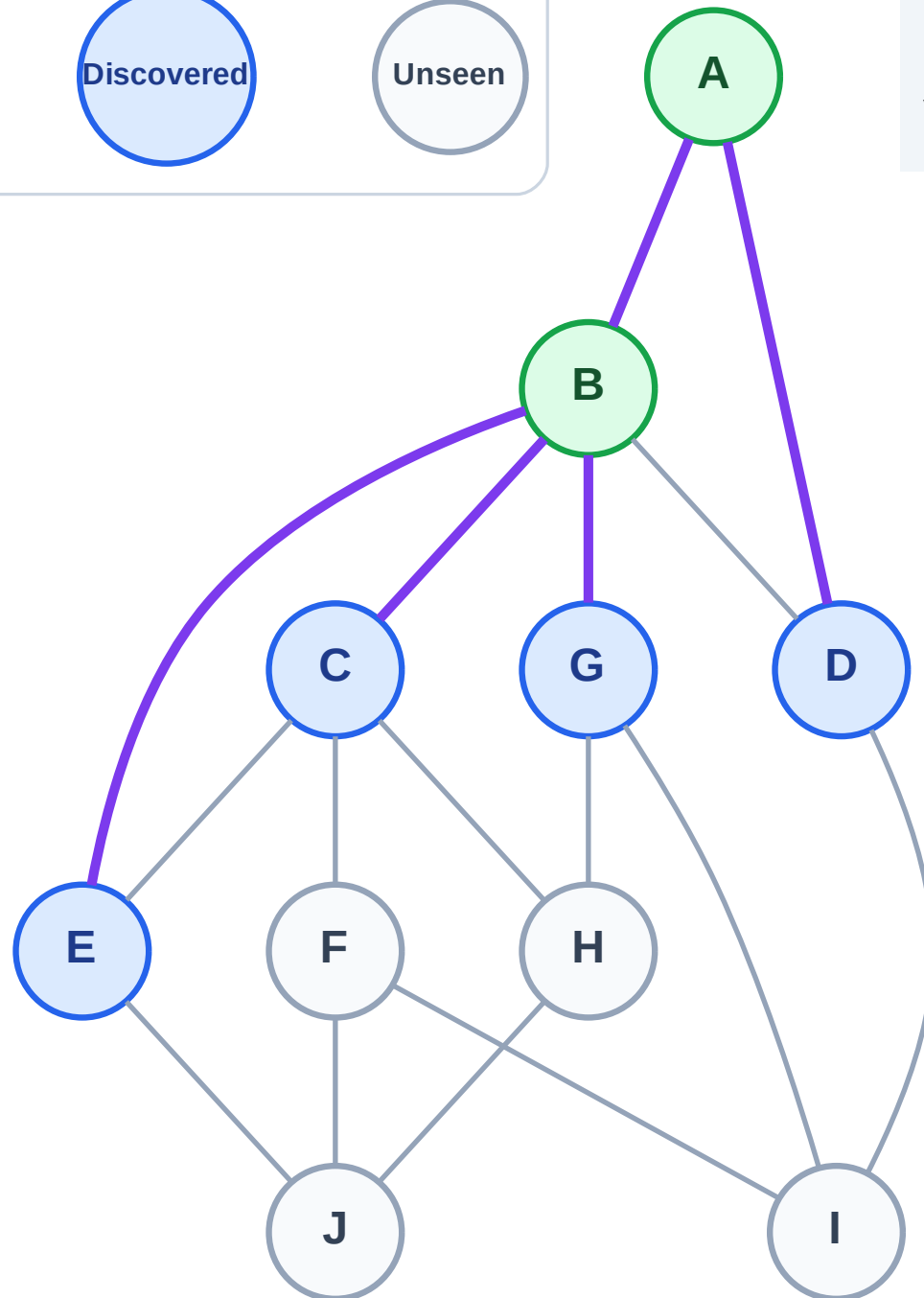
Step 5: Discover C, E, G from B

Legend



Queue: D → C → E → G

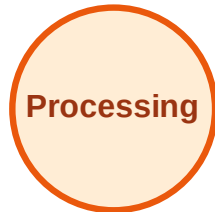
Visited: A, B



BFS Traversal

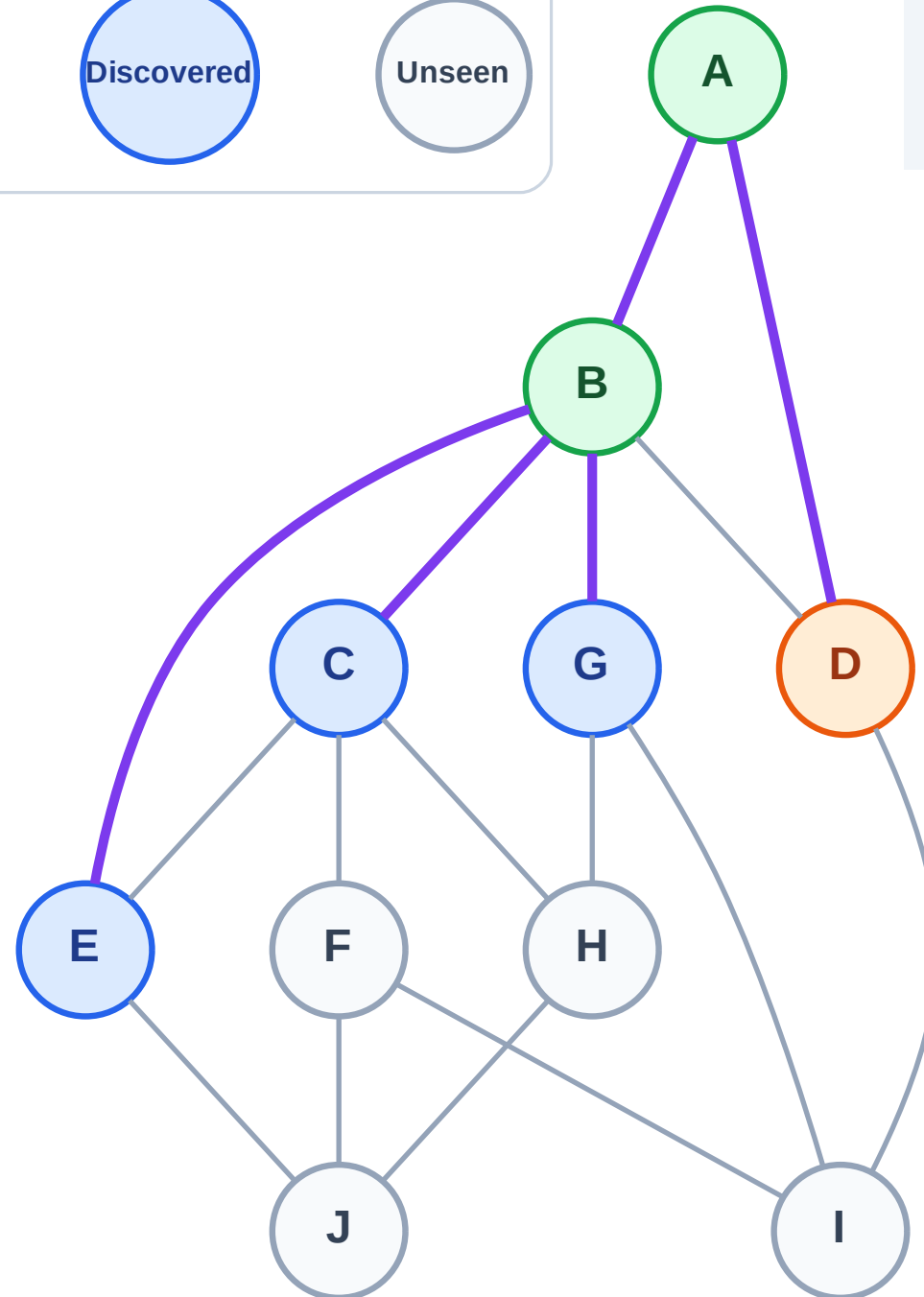
Step 6: Process node D

Legend



Queue: C → E → G

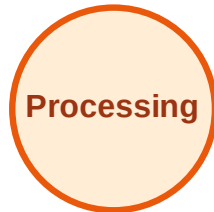
Visited: A, B



BFS Traversal

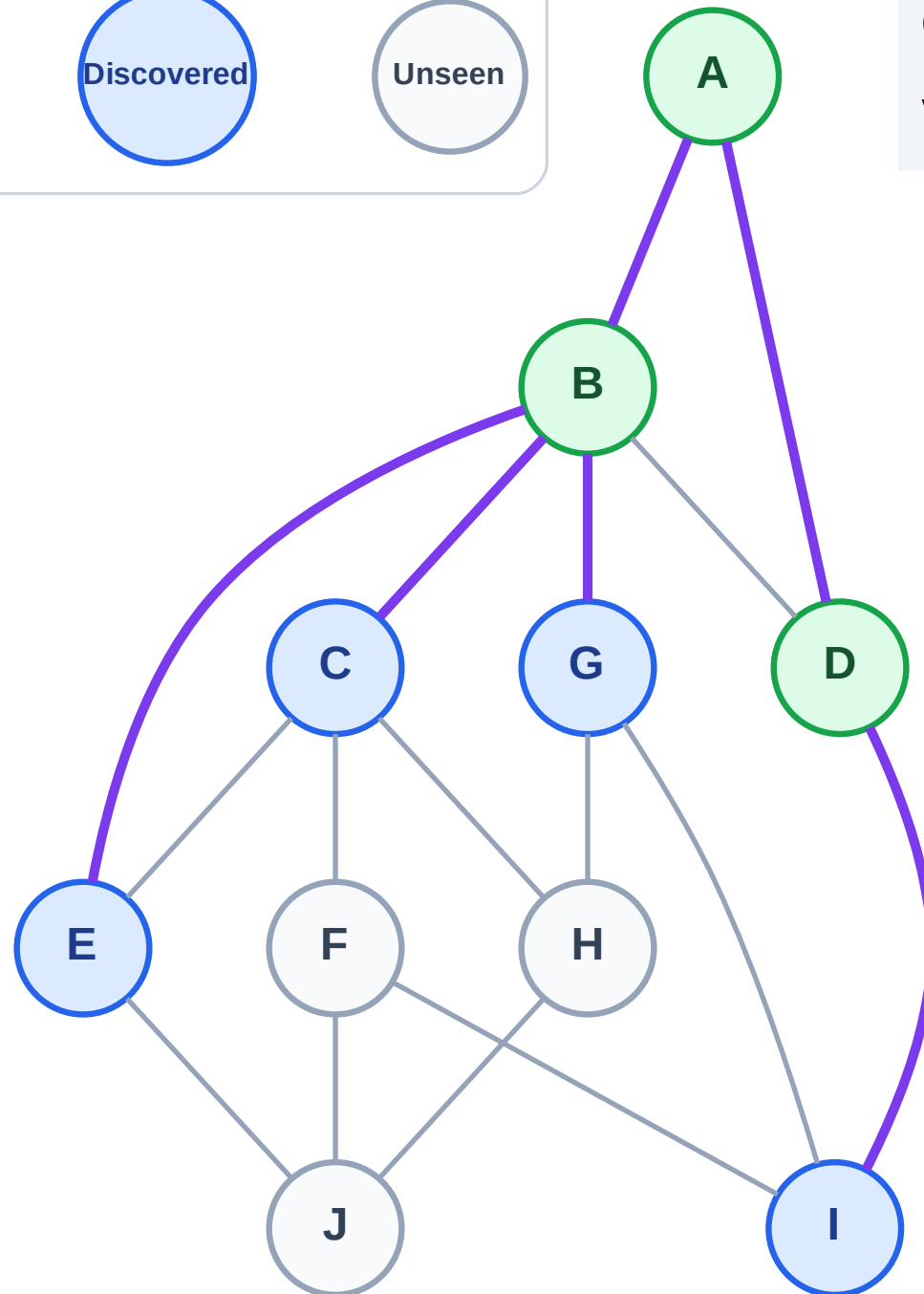
Step 7: Discover I from D

Legend



Queue: C → E → G → I

Visited: A, B, D



BFS Traversal

Step 8: Process node C

Legend

Complete

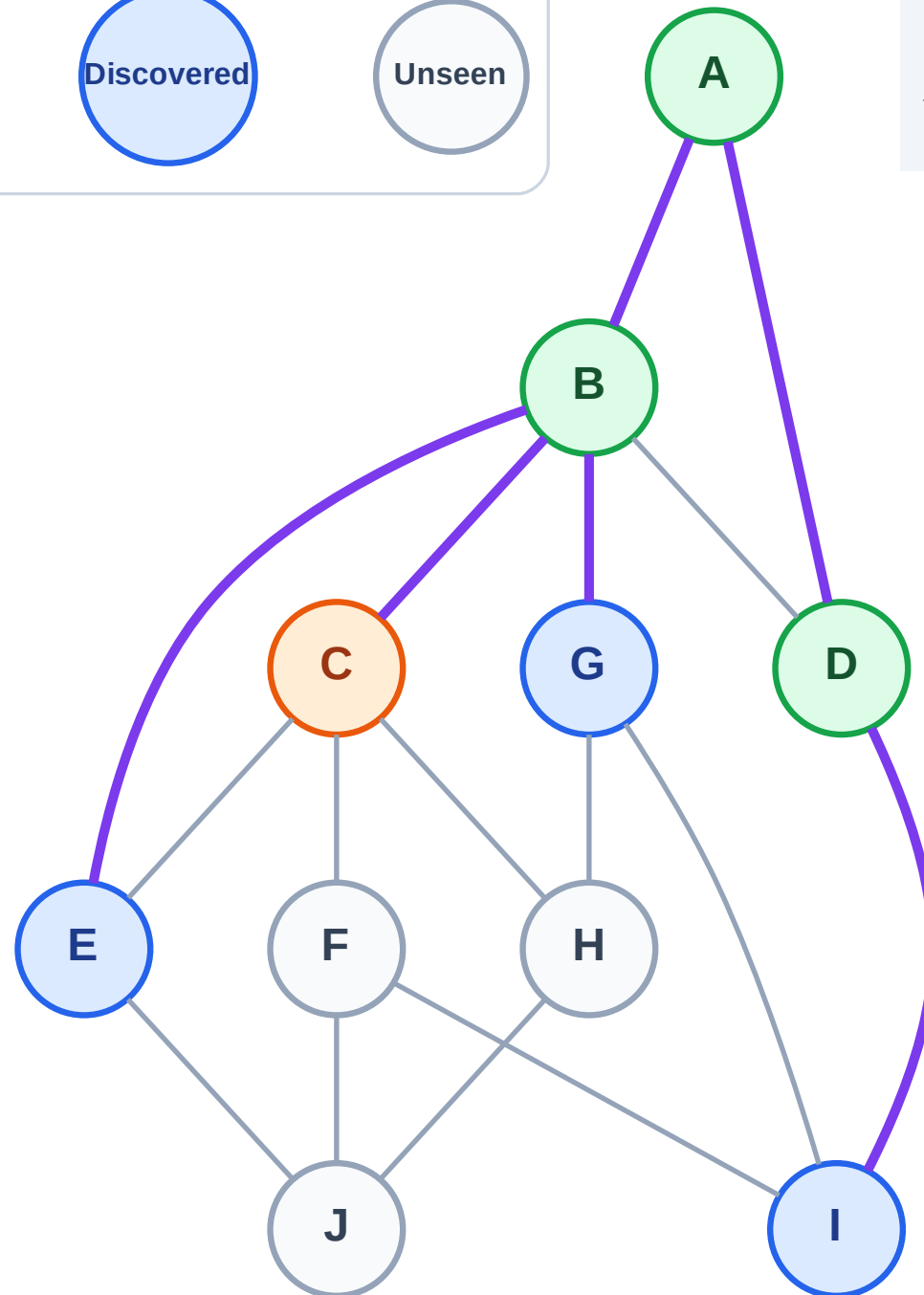
Processing

Discovered

Unseen

Queue: E → G → I

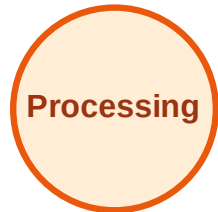
Visited: A, B, D



BFS Traversal

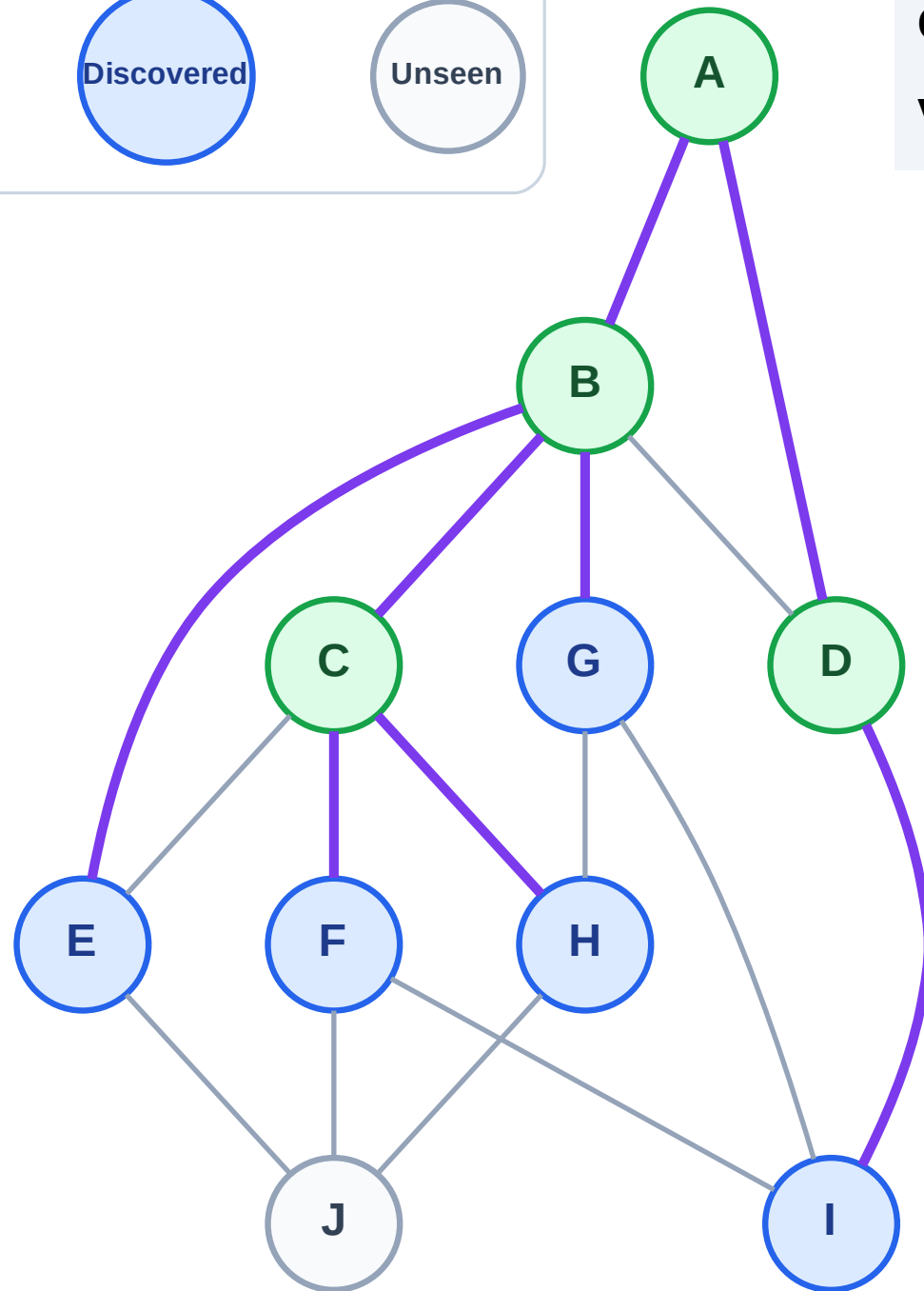
Step 9: Discover F, H from C

Legend



Queue: E → G → I → F → H

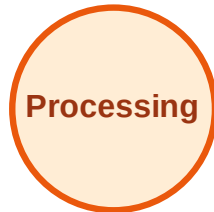
Visited: A, B, C, D



BFS Traversal

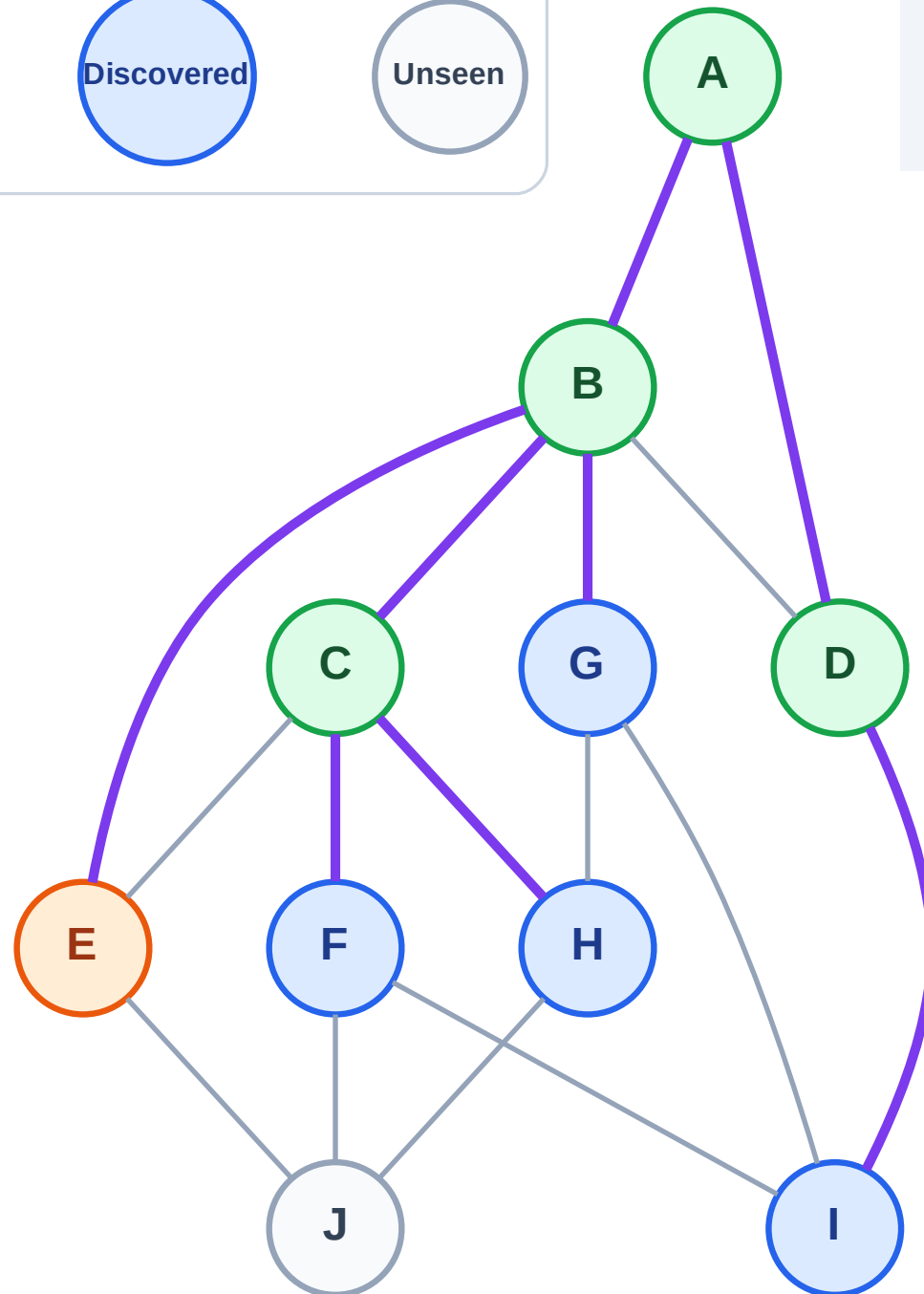
Step 10: Process node E

Legend



Queue: G → I → F → H

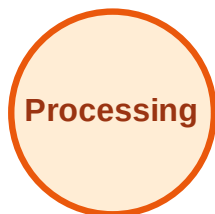
Visited: A, B, C, D



BFS Traversal

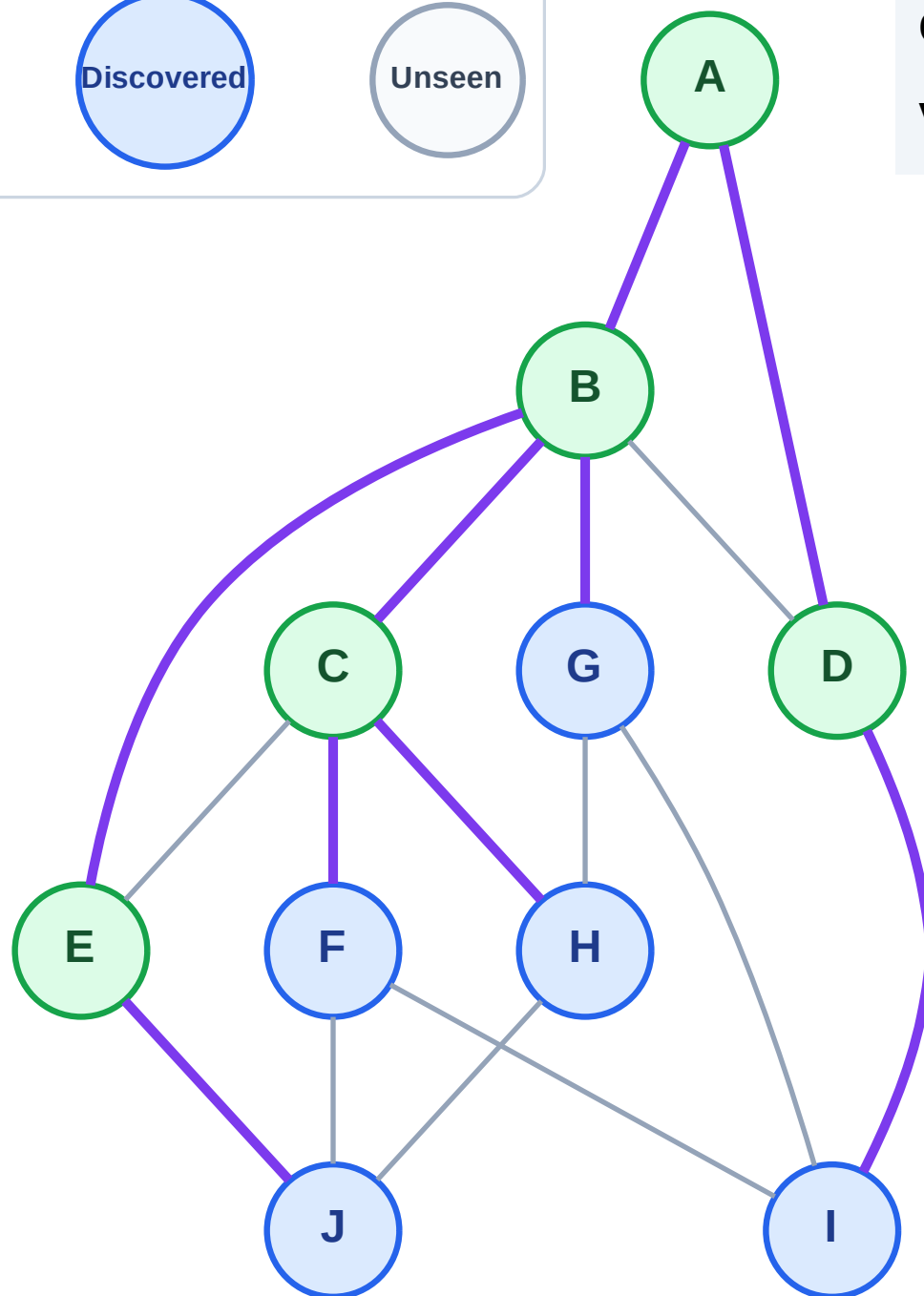
Step 11: Discover J from E

Legend



Queue: G → I → F → H → J

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

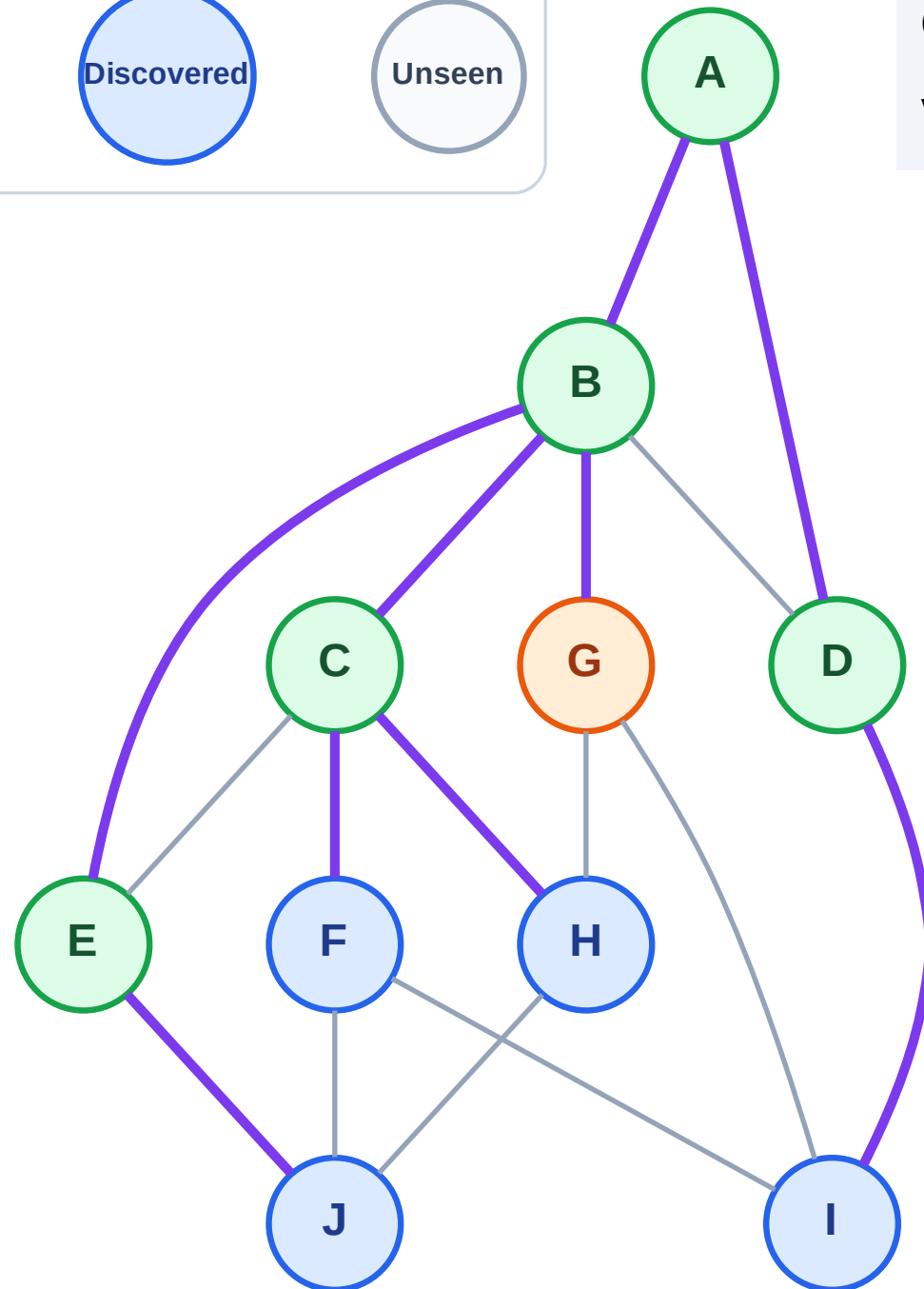
Processing

Discovered

Unseen

Queue: I → F → H → J

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

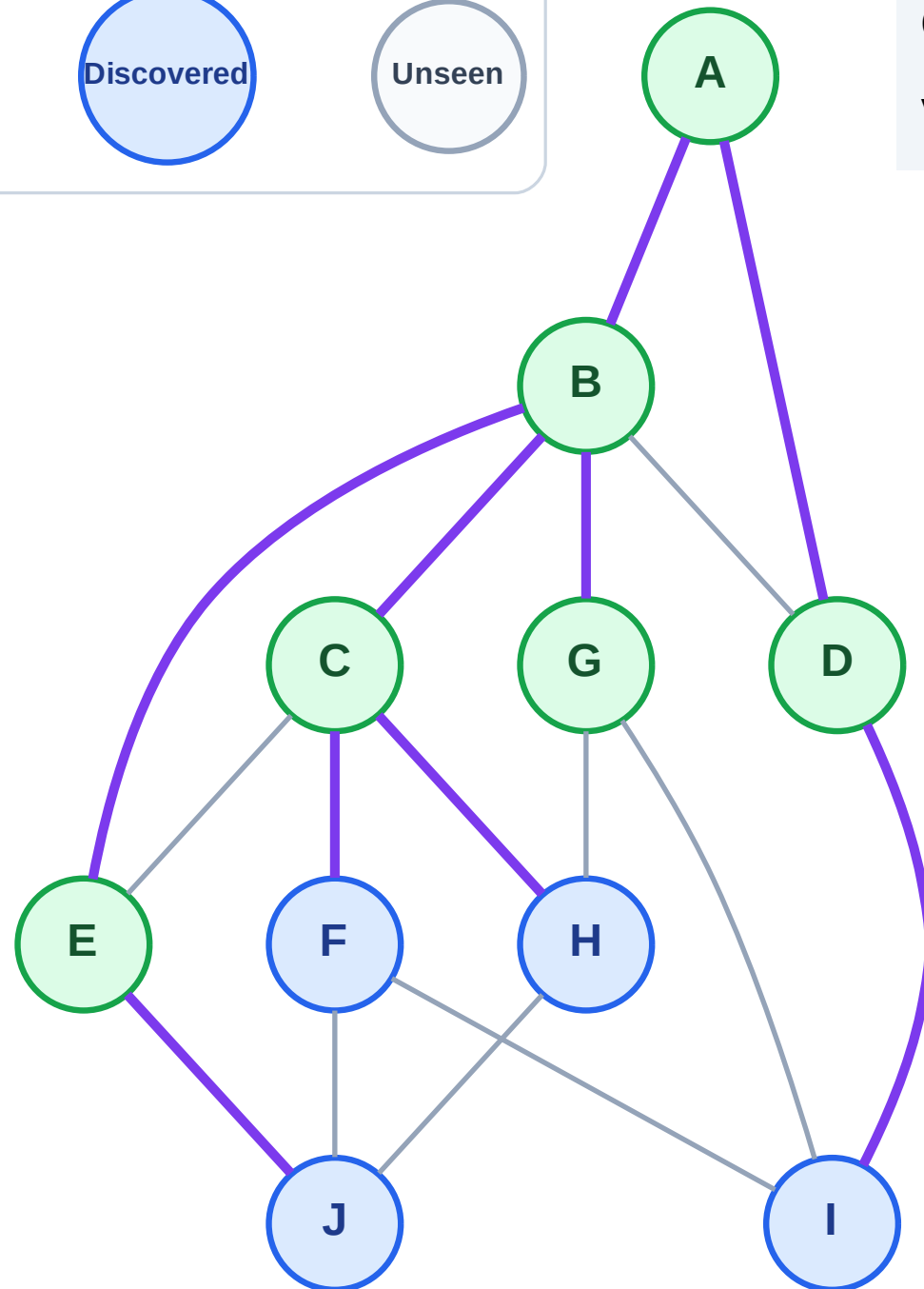
Processing

Discovered

Unseen

Queue: I → F → H → J

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node I

Legend

Complete

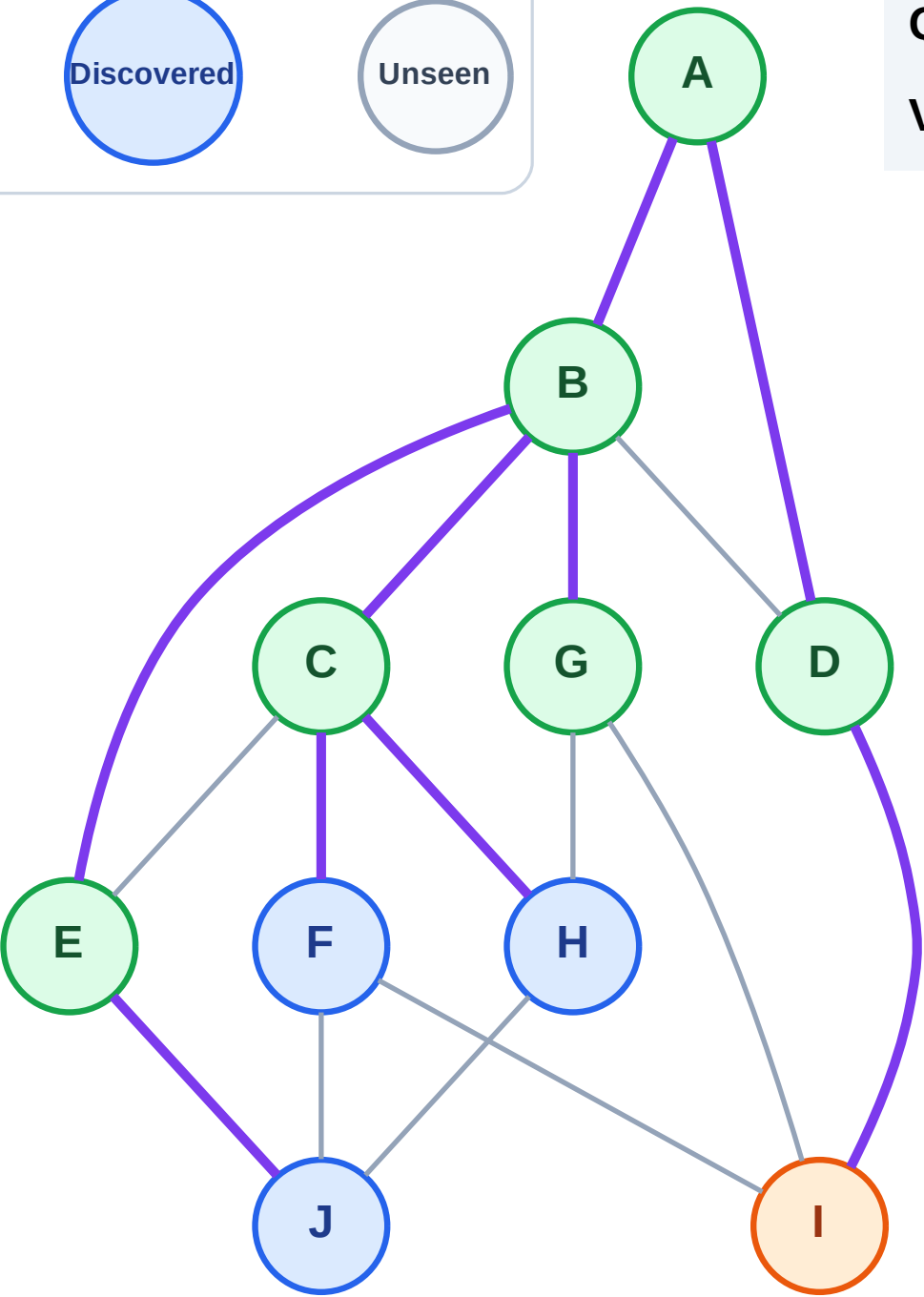
Processing

Discovered

Unseen

Queue: F → H → J

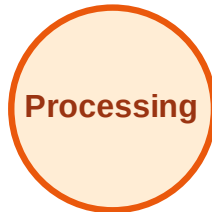
Visited: A, B, C, D, E, G



BFS Traversal

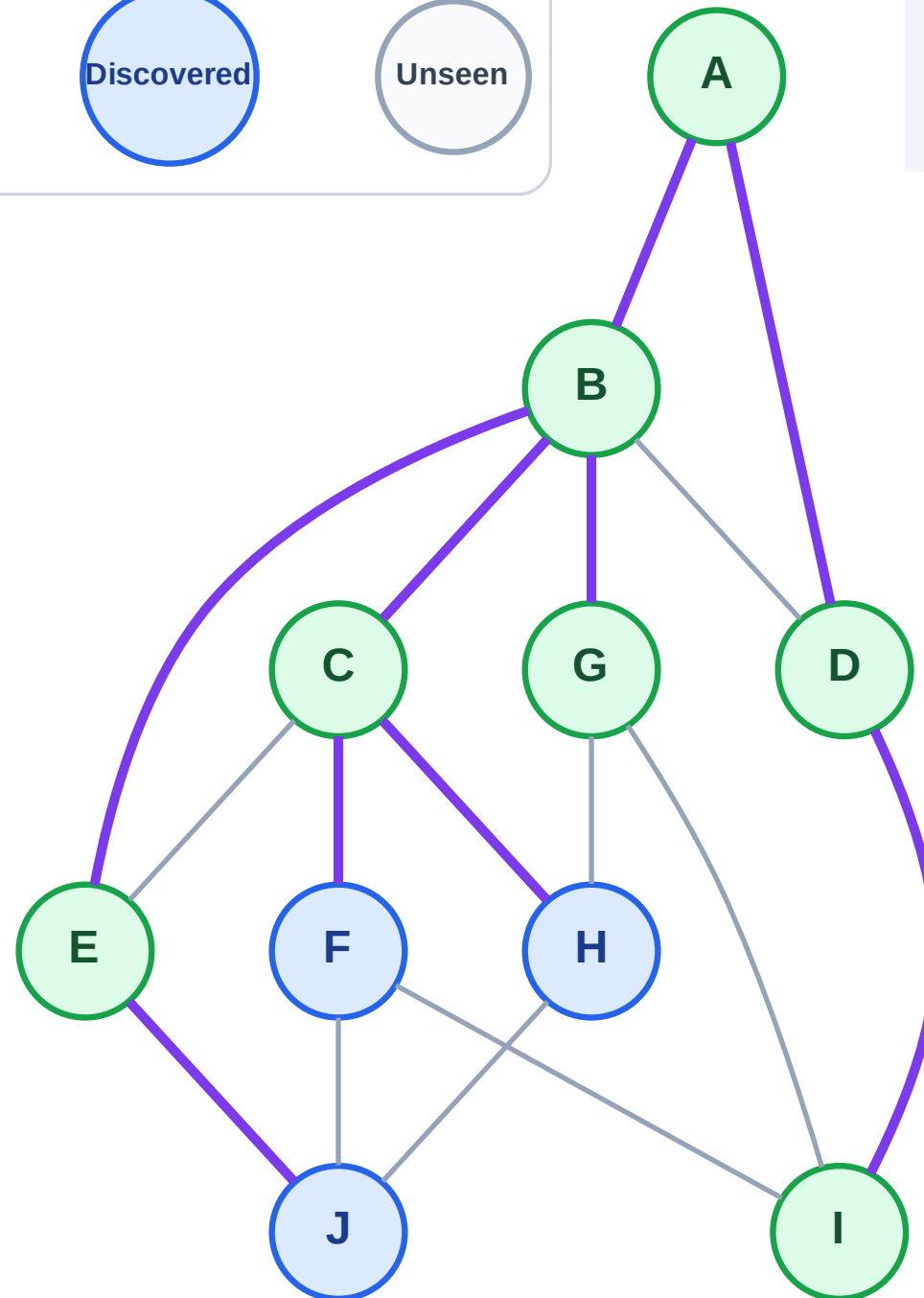
Step 15: No new neighbors from I

Legend



Queue: F → H → J

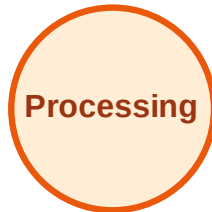
Visited: A, B, C, D, E, G, I



BFS Traversal

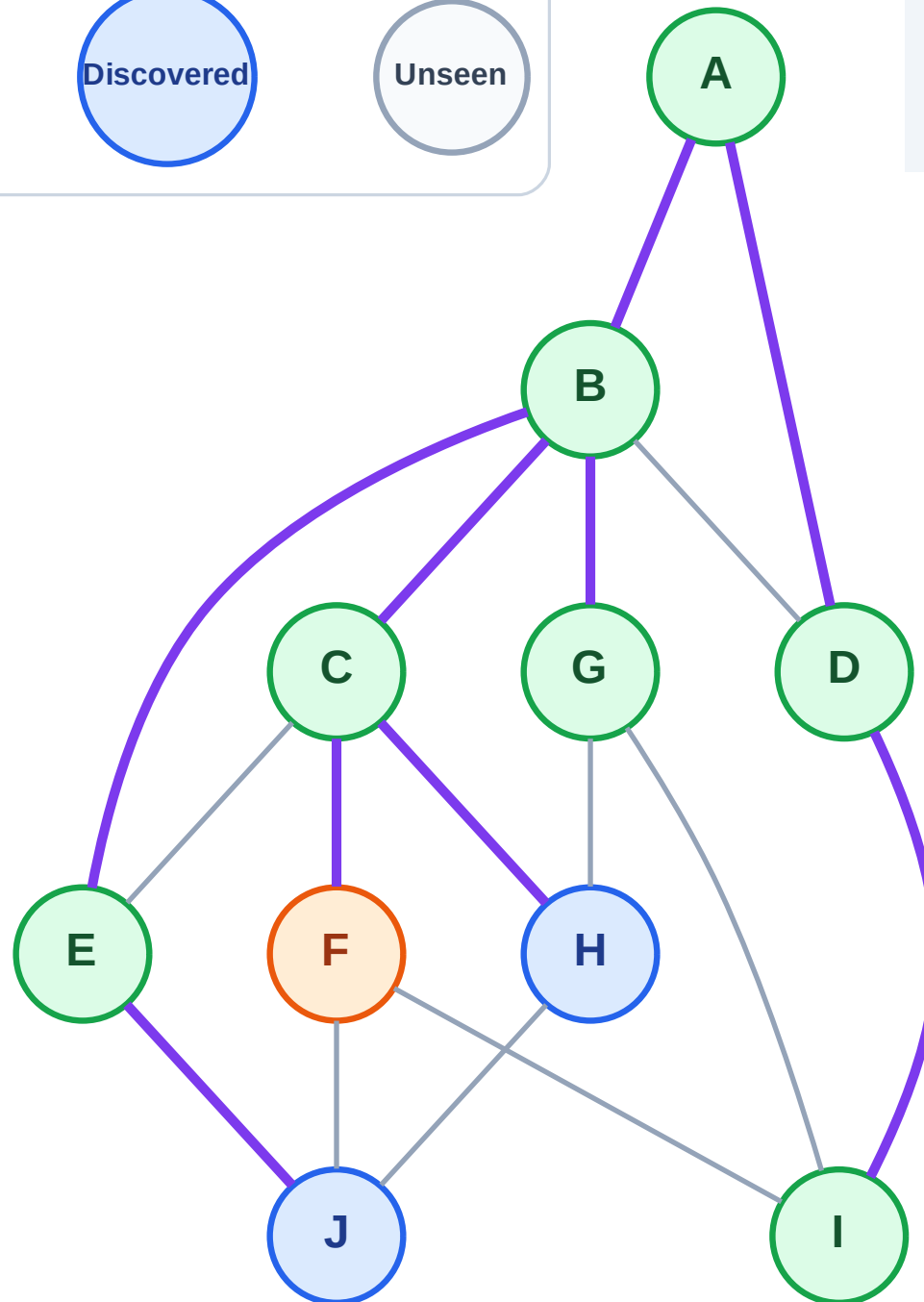
Step 16: Process node F

Legend



Queue: H → J

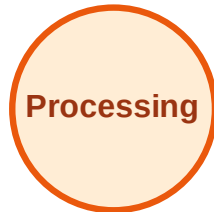
Visited: A, B, C, D, E, G, I



BFS Traversal

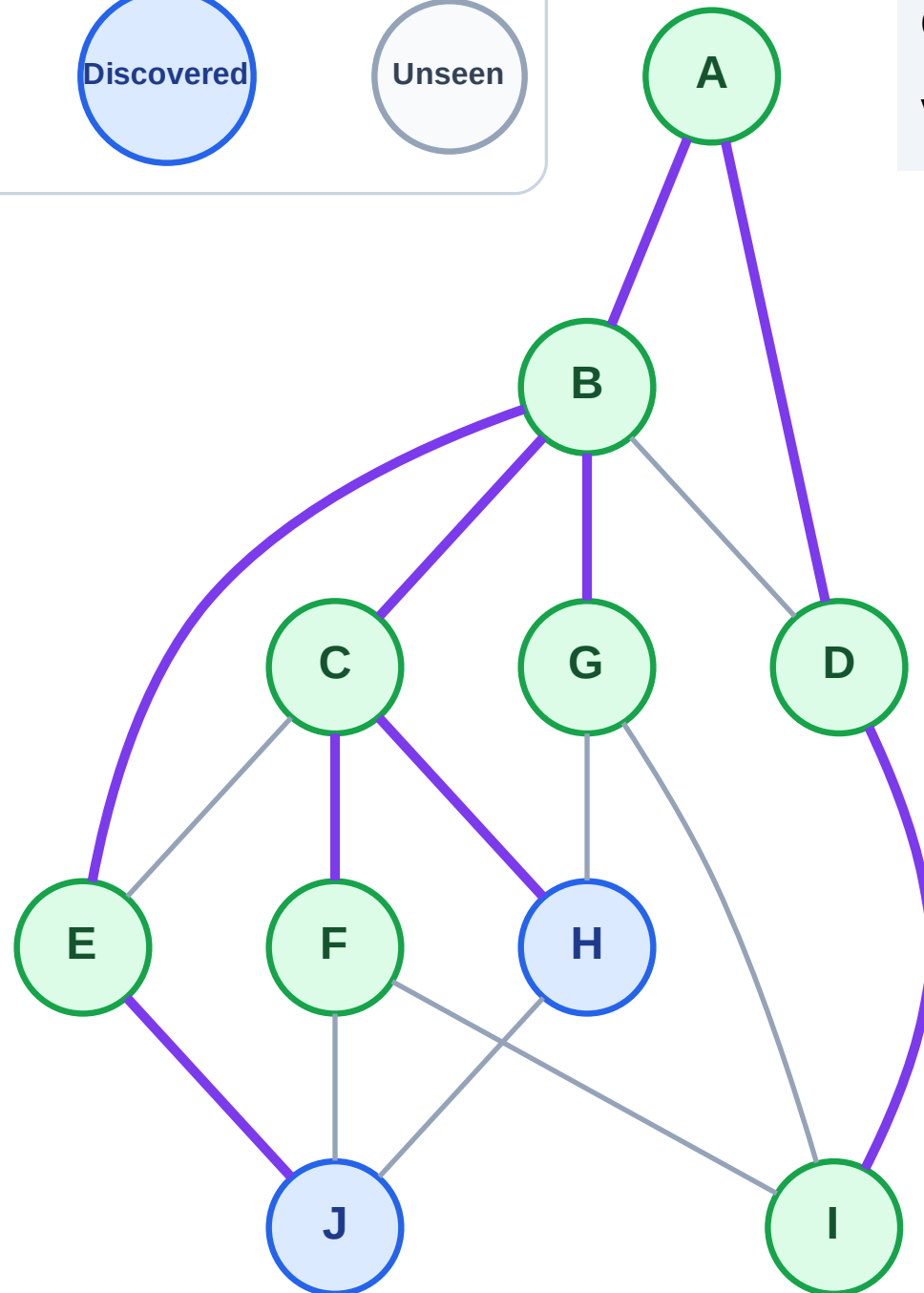
Step 17: No new neighbors from F

Legend



Queue: H → J

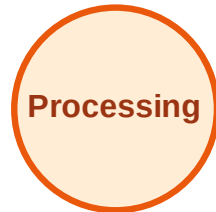
Visited: A, B, C, D, E, F, G, I



BFS Traversal

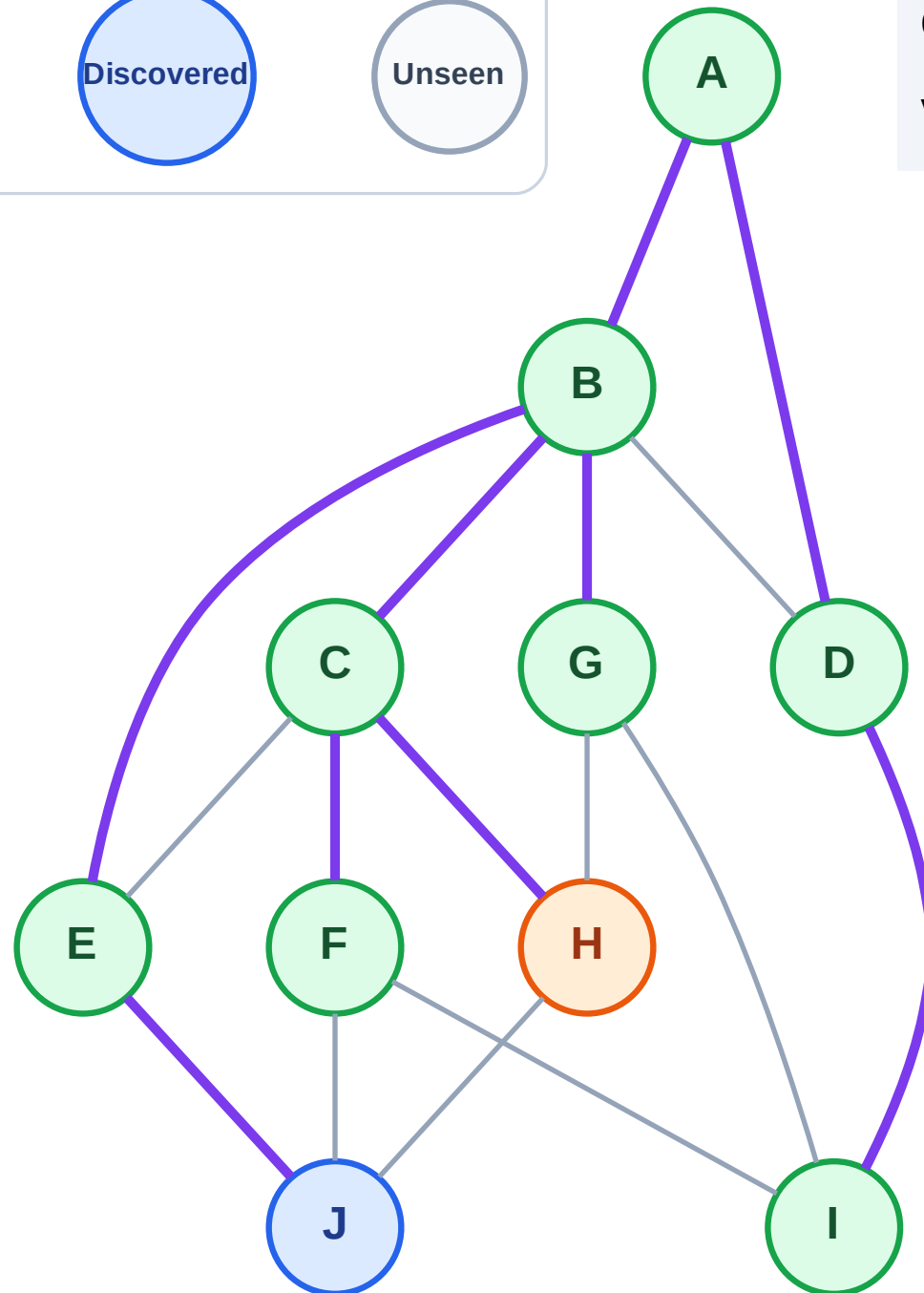
Step 18: Process node H

Legend



Queue: J

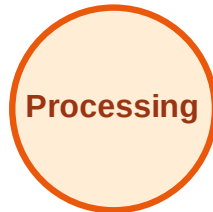
Visited: A, B, C, D, E, F, G, I



BFS Traversal

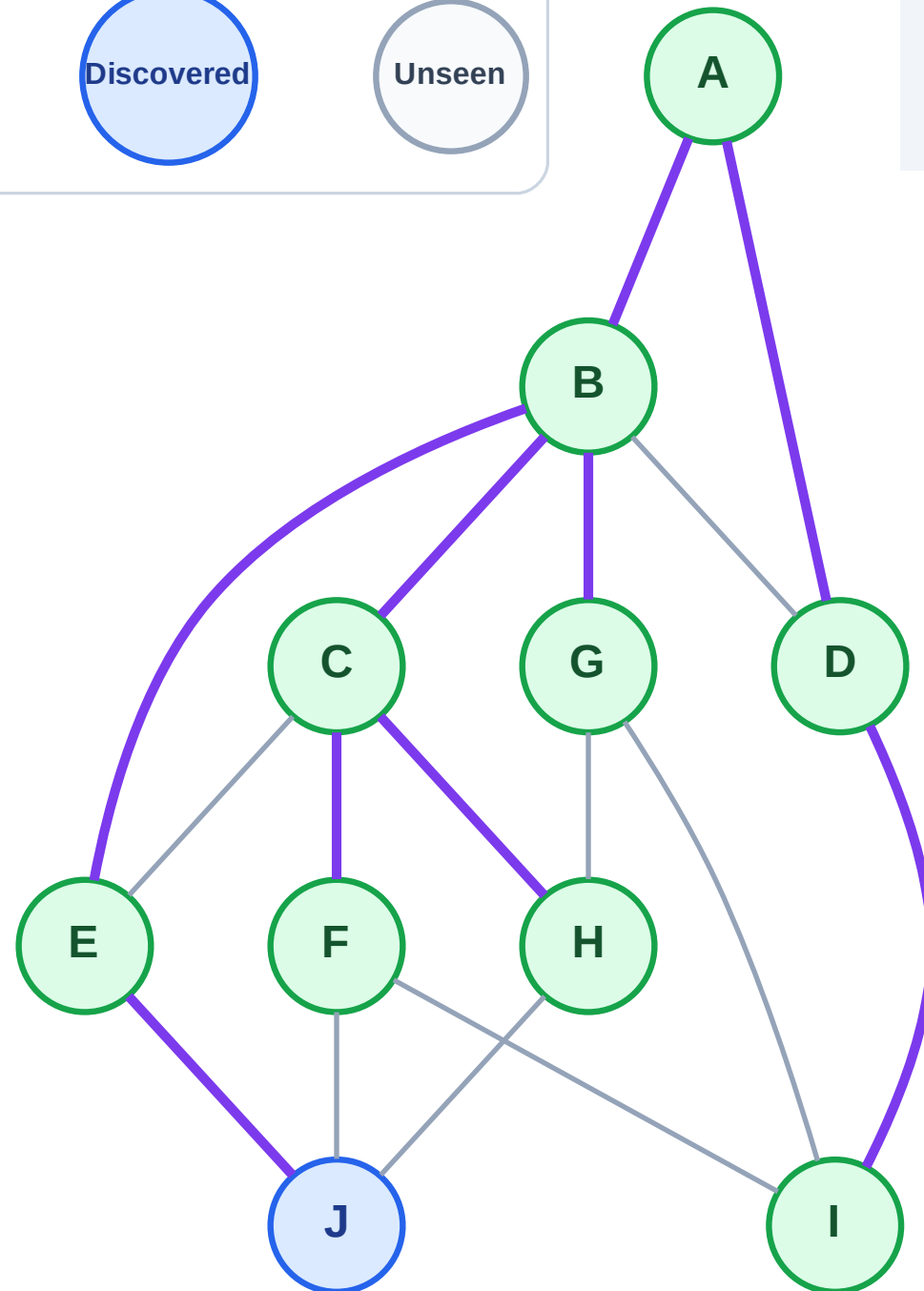
Step 19: No new neighbors from H

Legend



Queue: J

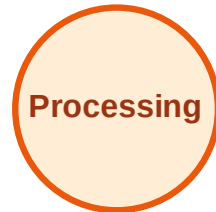
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

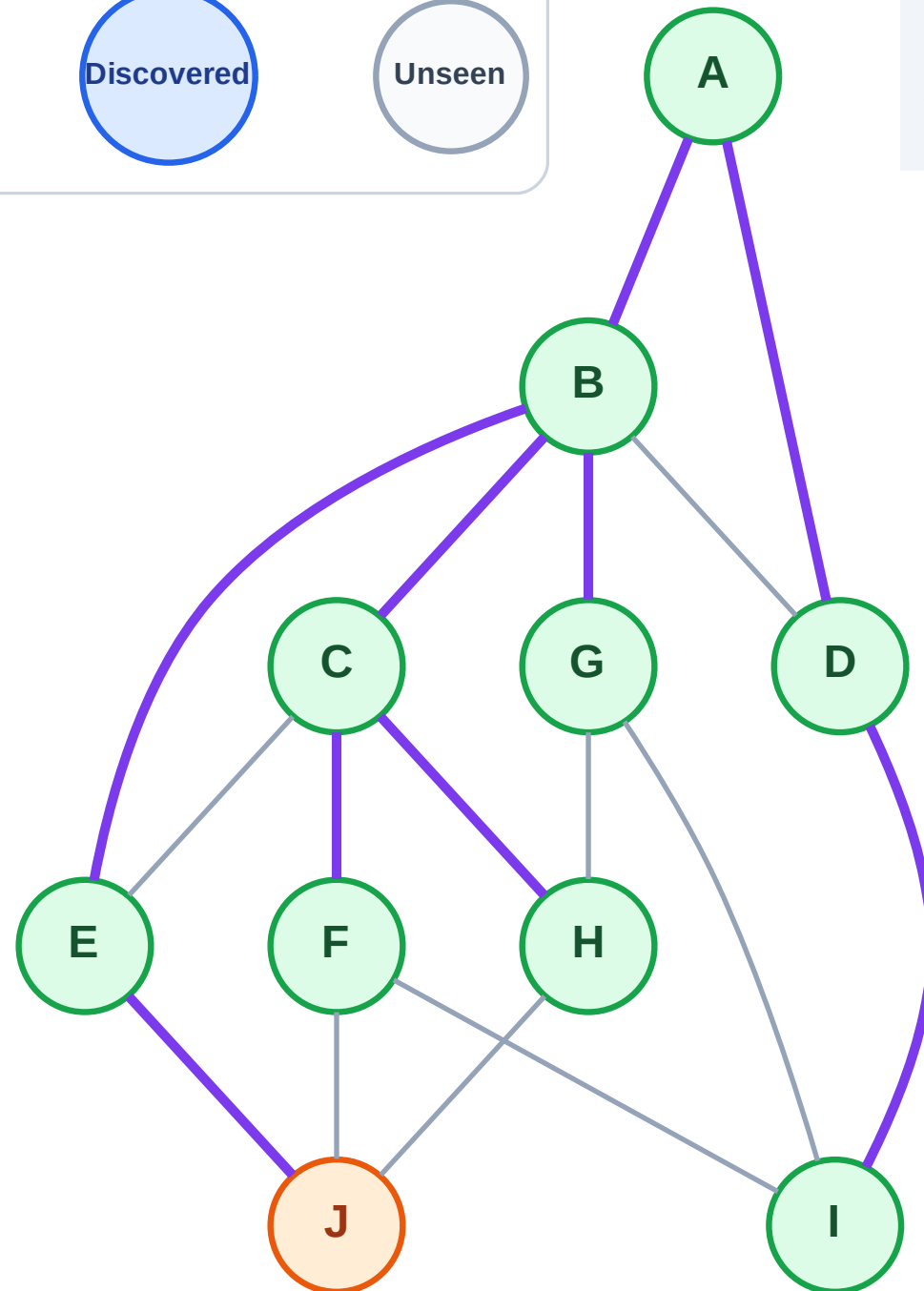
Step 20: Process node J

Legend



Queue: (empty)

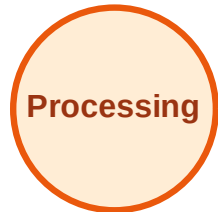
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

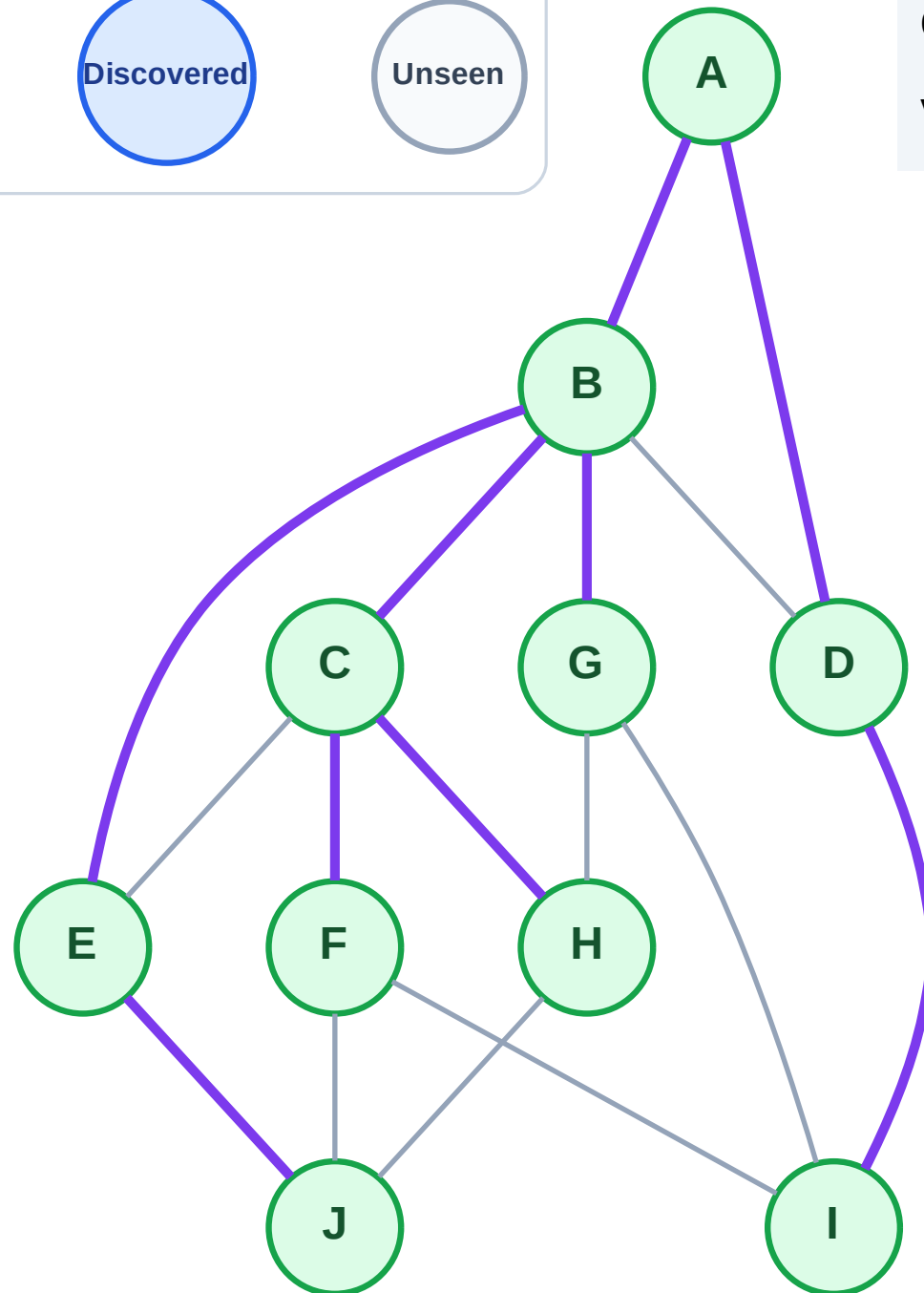
Step 21: No new neighbors from J

Legend



Queue: (empty)

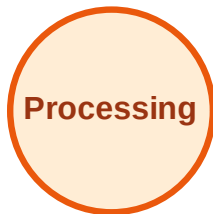
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

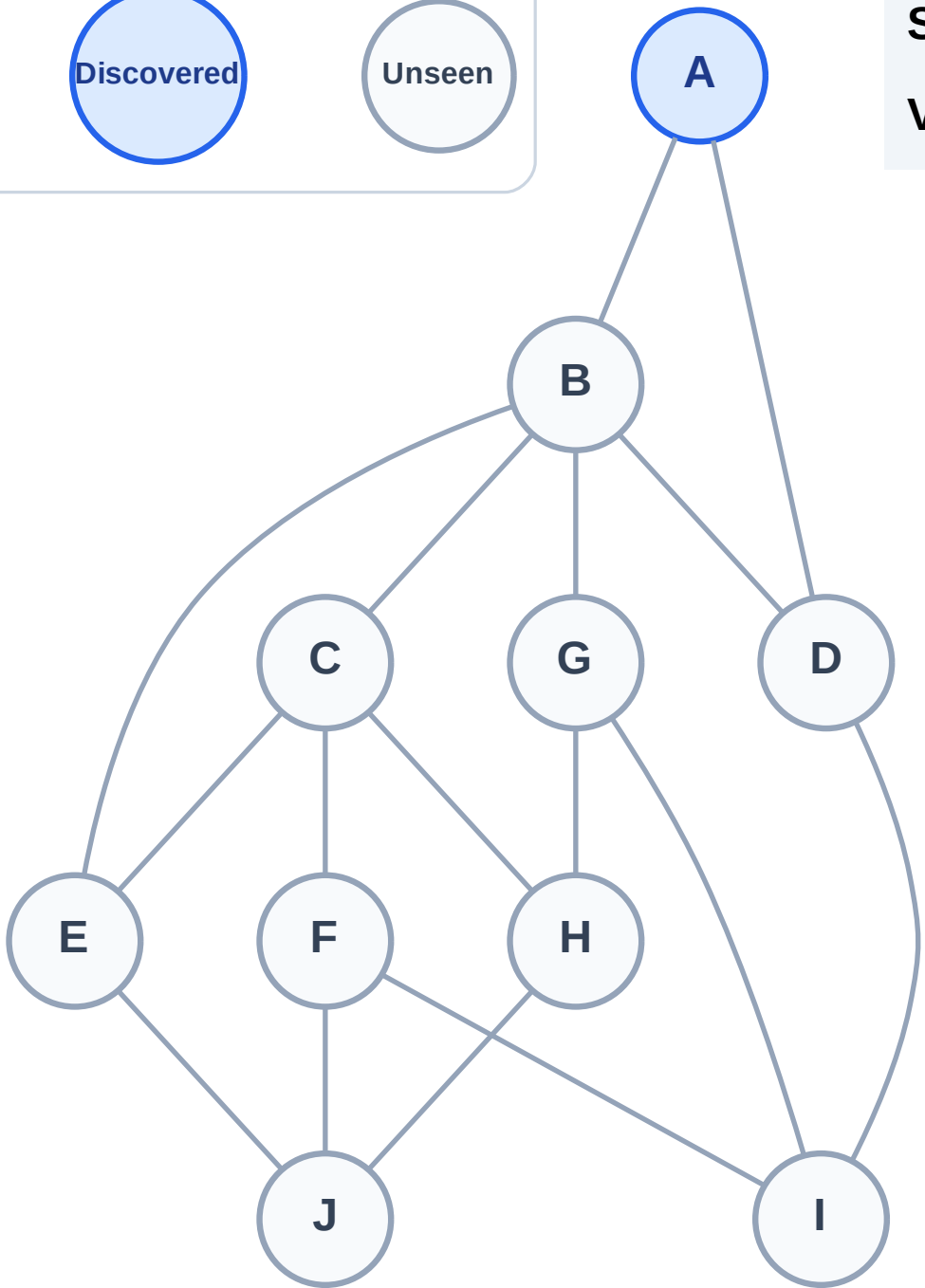
Step 1: Start at A

Legend



Stack (top at right): A

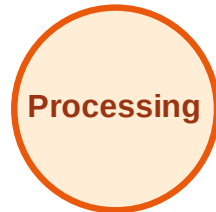
Visited: (none)



DFS Traversal

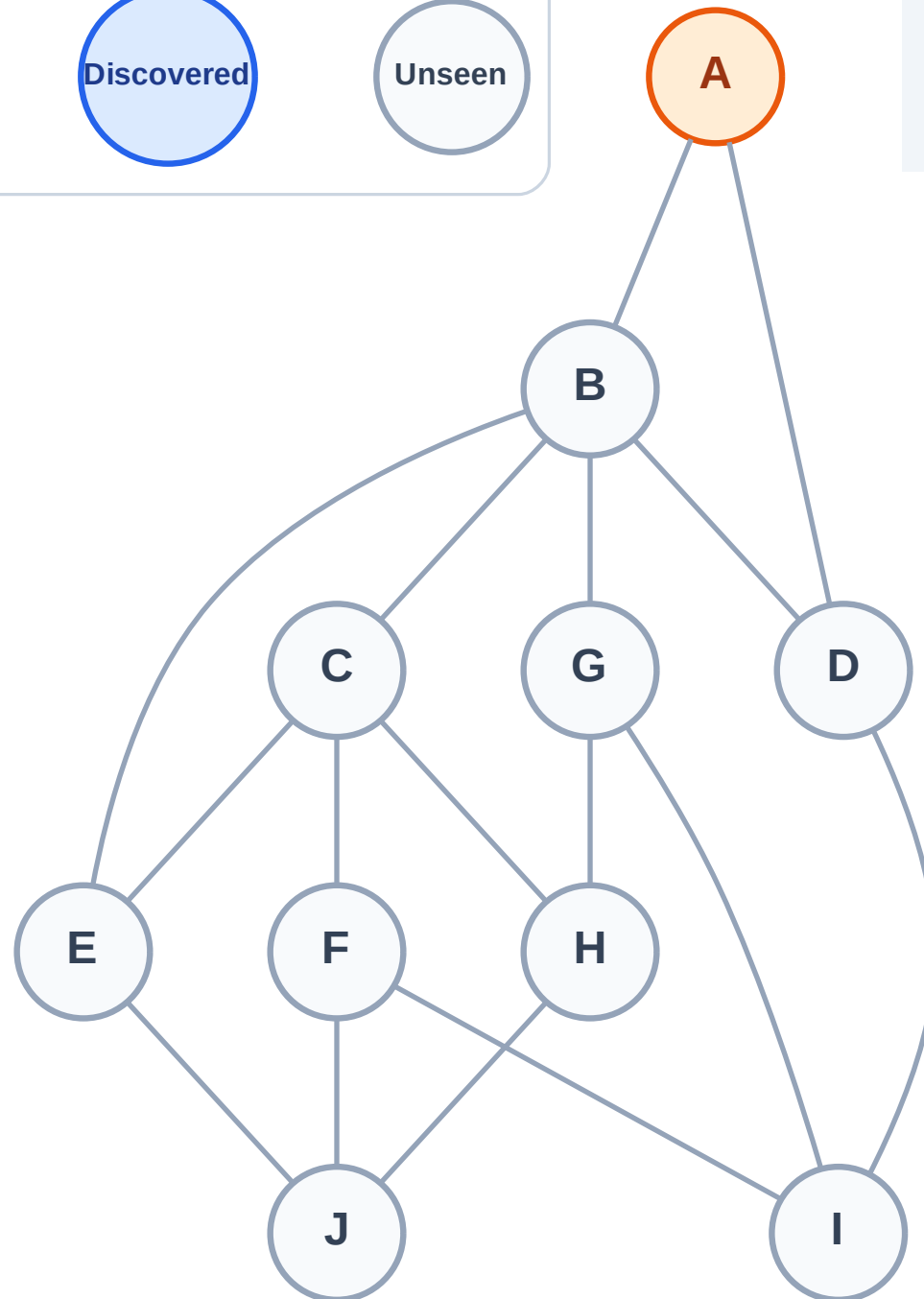
Step 2: Process node A

Legend



Stack (top at right): (empty)

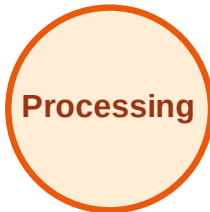
Visited: (none)



DFS Traversal

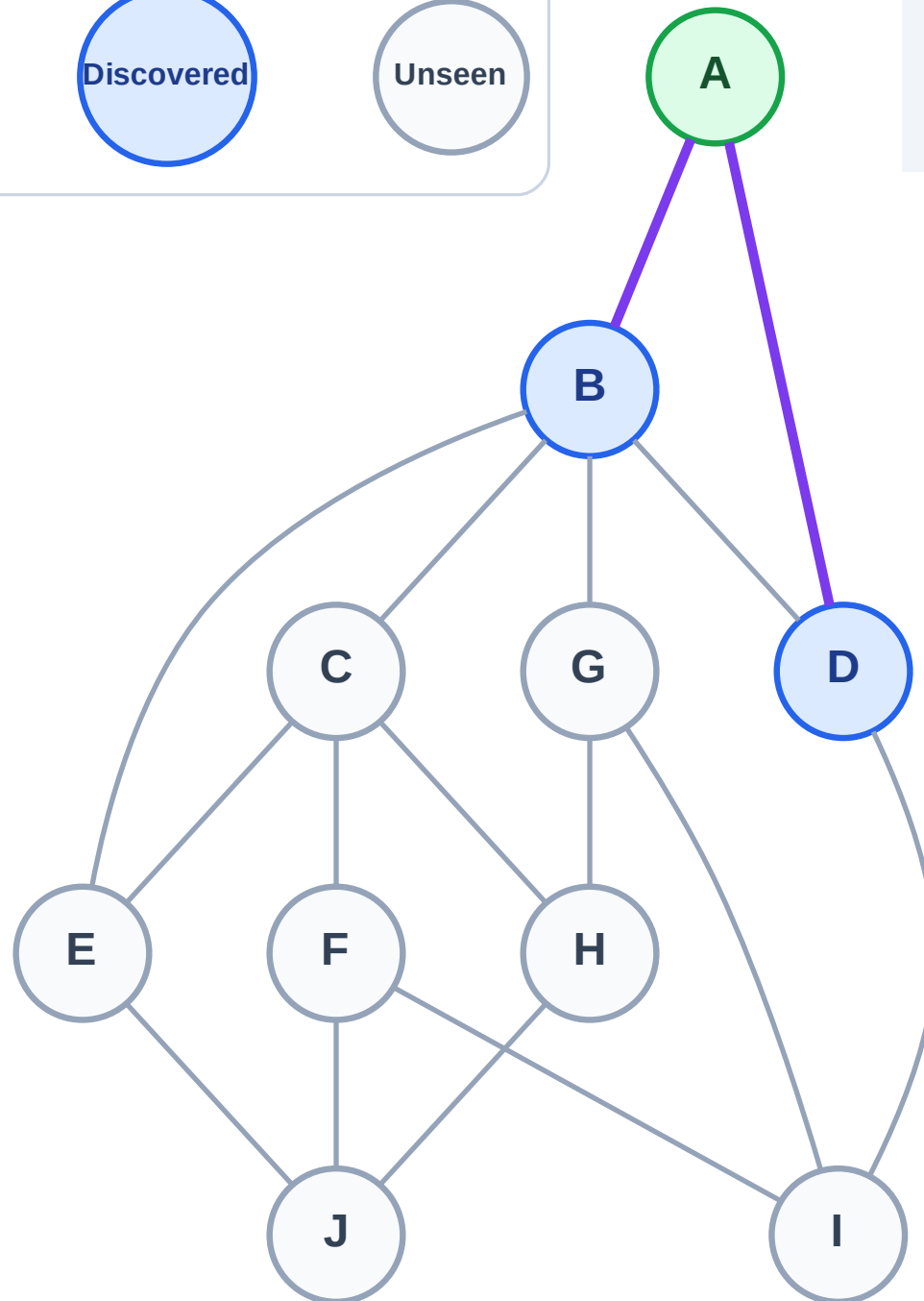
Step 3: Discover D, B from A

Legend



Stack (top at right): D | B

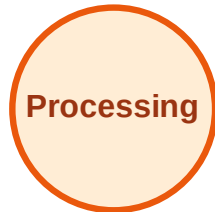
Visited: A



DFS Traversal

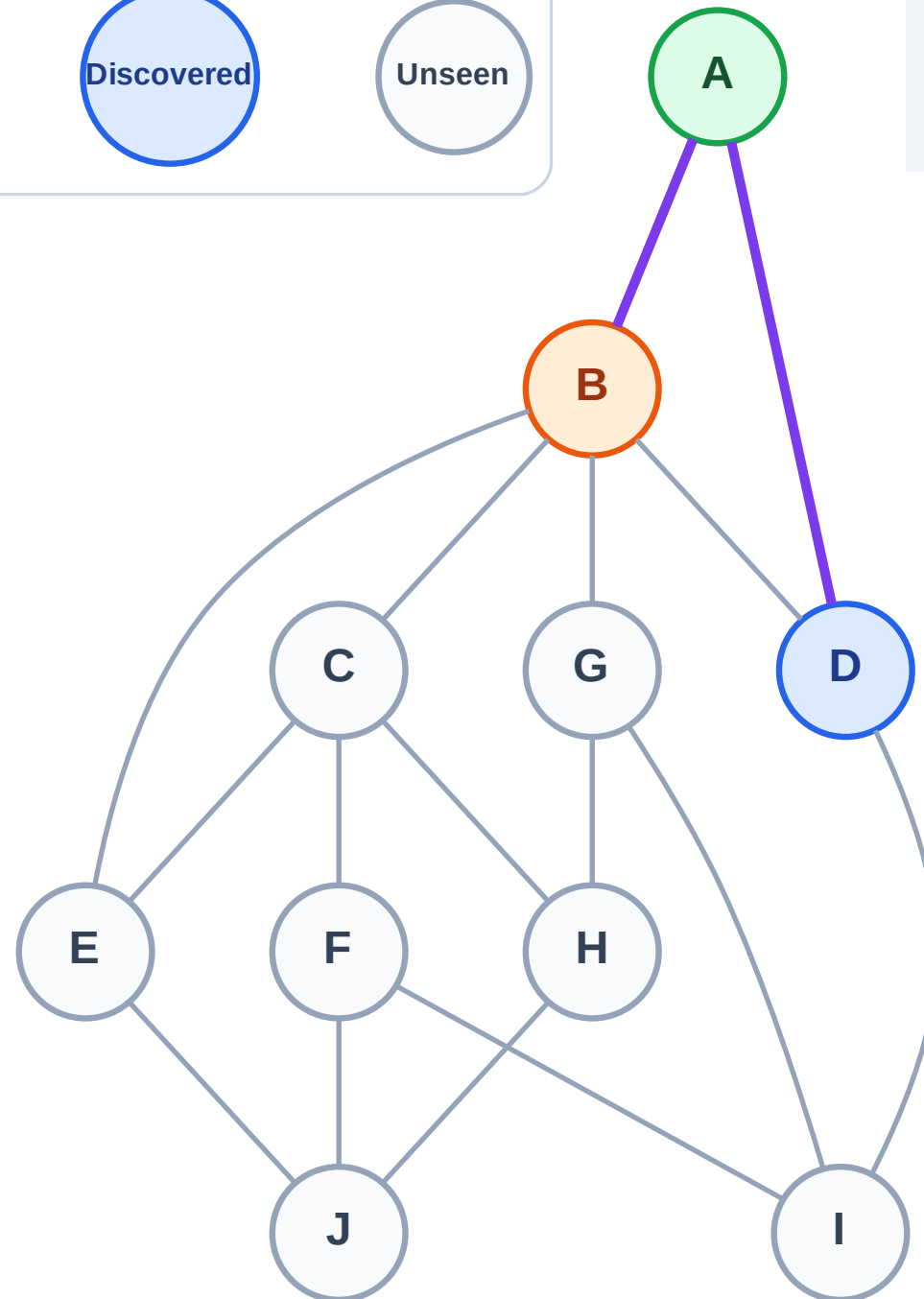
Step 4: Process node B

Legend



Stack (top at right): D

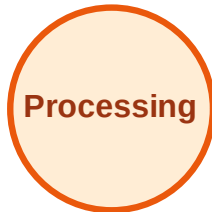
Visited: A



DFS Traversal

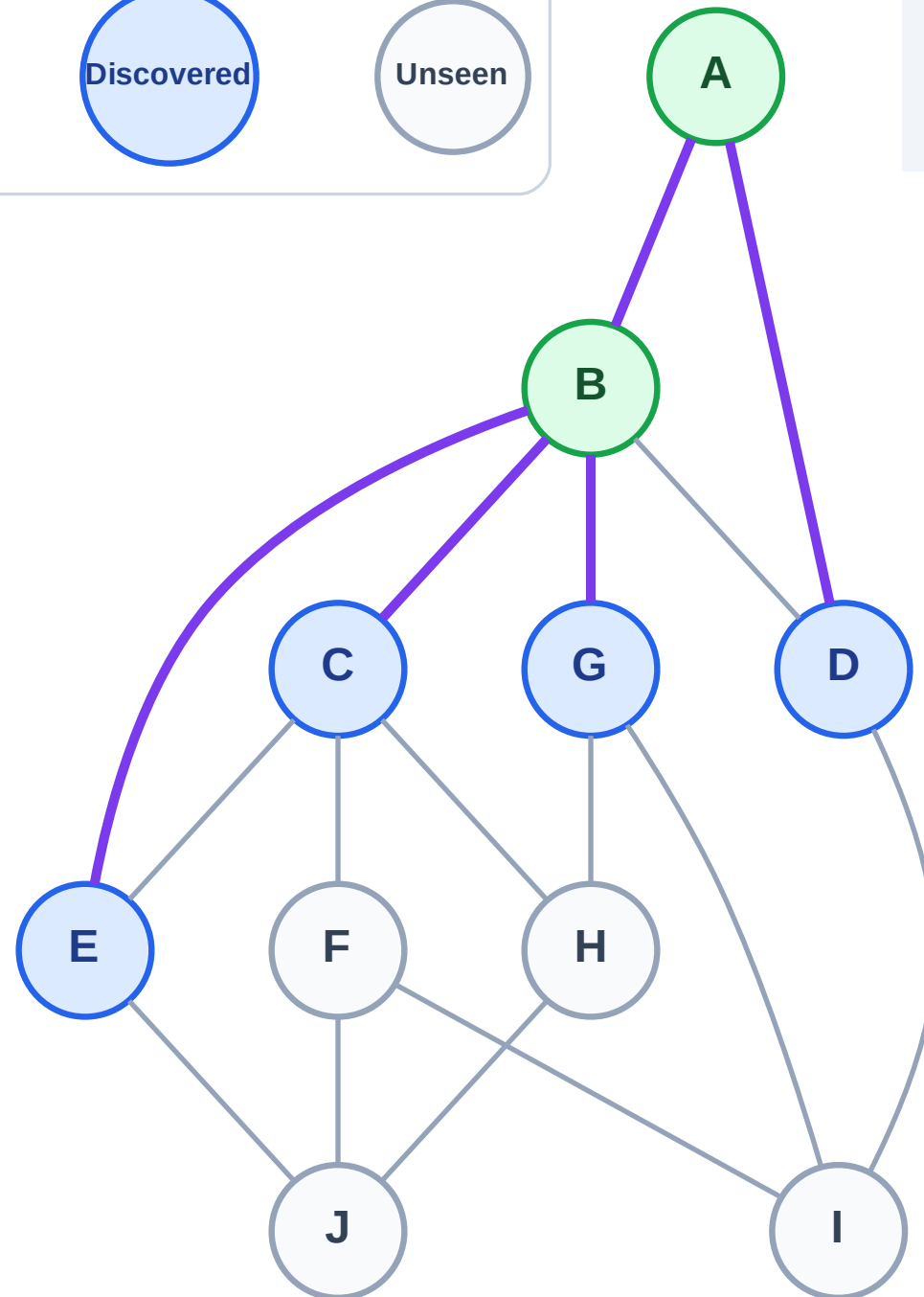
Step 5: Discover G, E, C from B

Legend



Stack (top at right): D | G | E | C

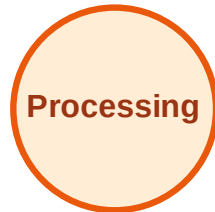
Visited: A, B



DFS Traversal

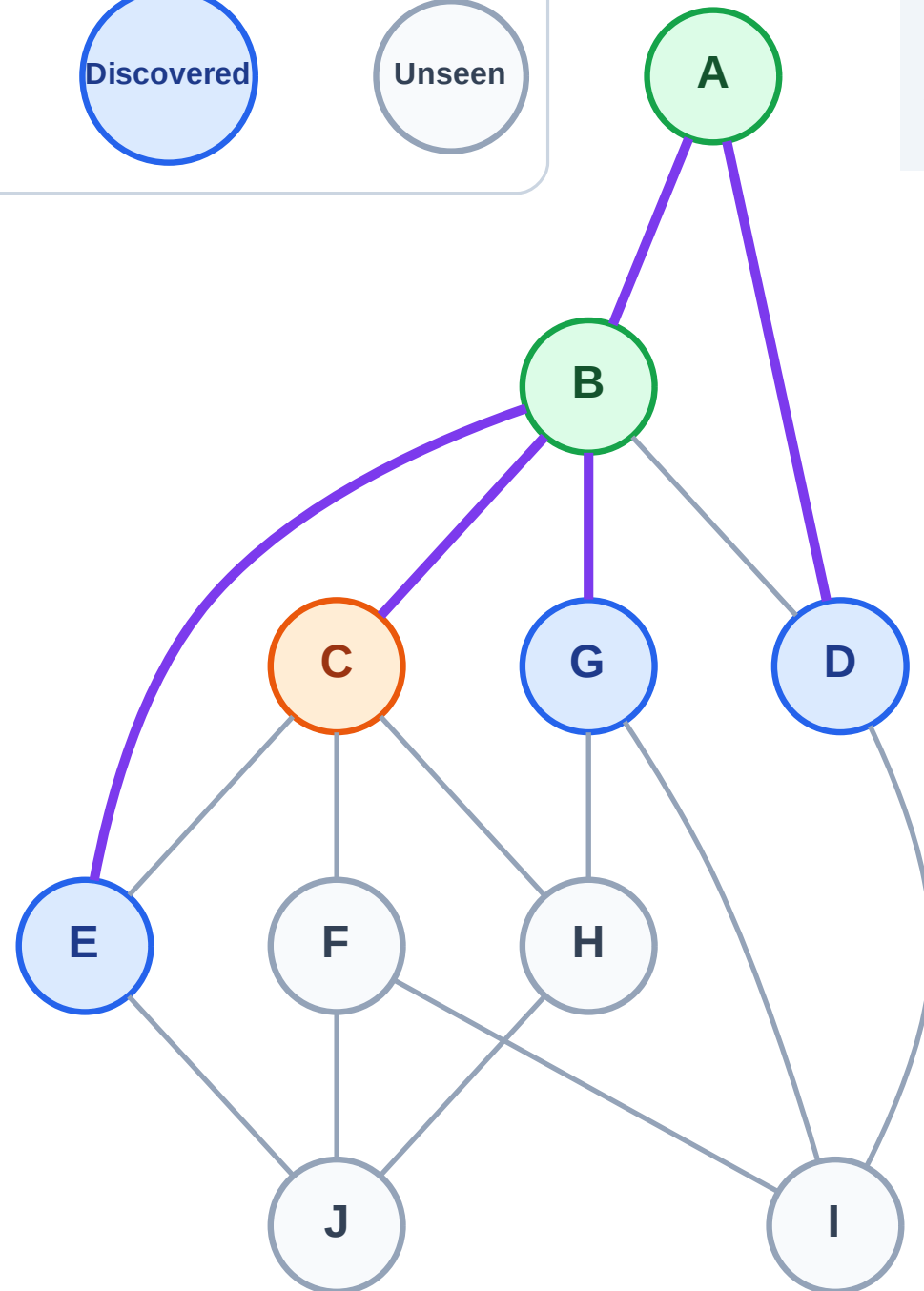
Step 6: Process node C

Legend



Stack (top at right): D | G | E

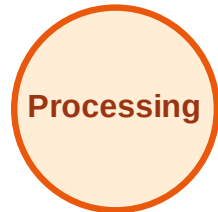
Visited: A, B



DFS Traversal

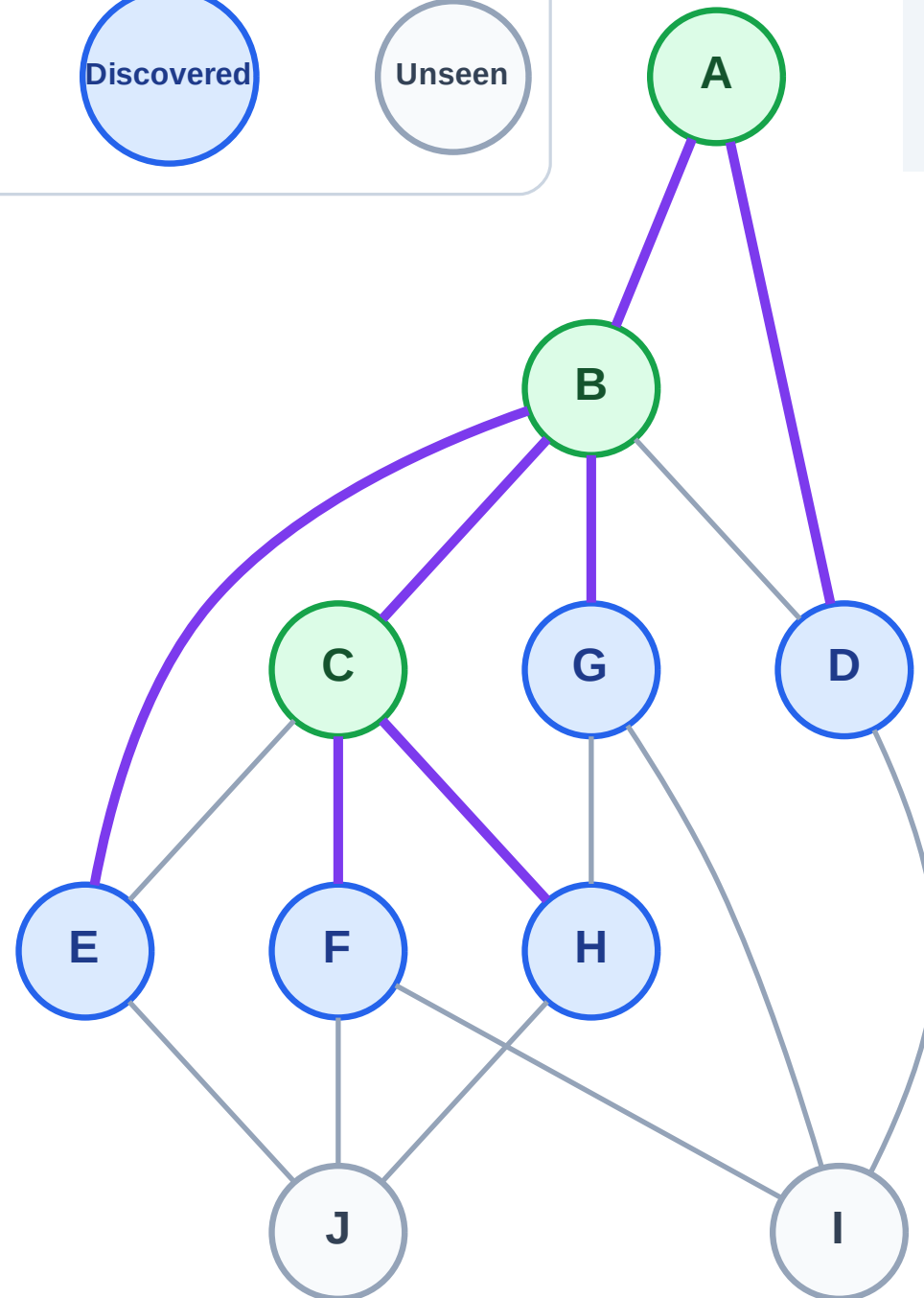
Step 7: Discover H, F from C

Legend



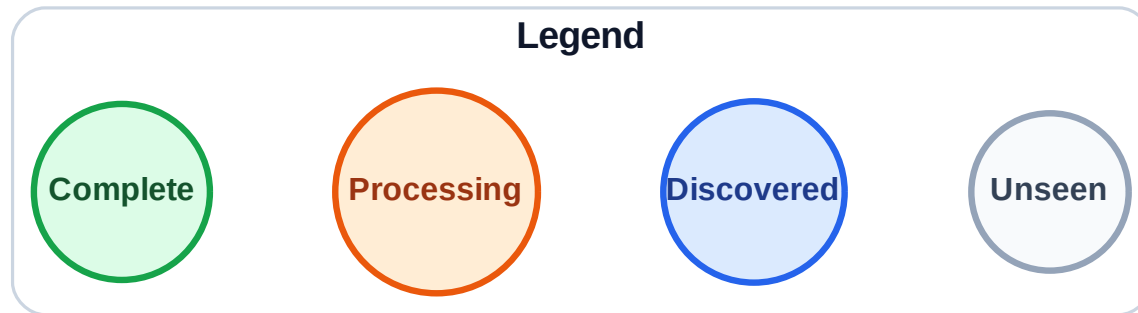
Stack (top at right): D | G | E | H | F

Visited: A, B, C



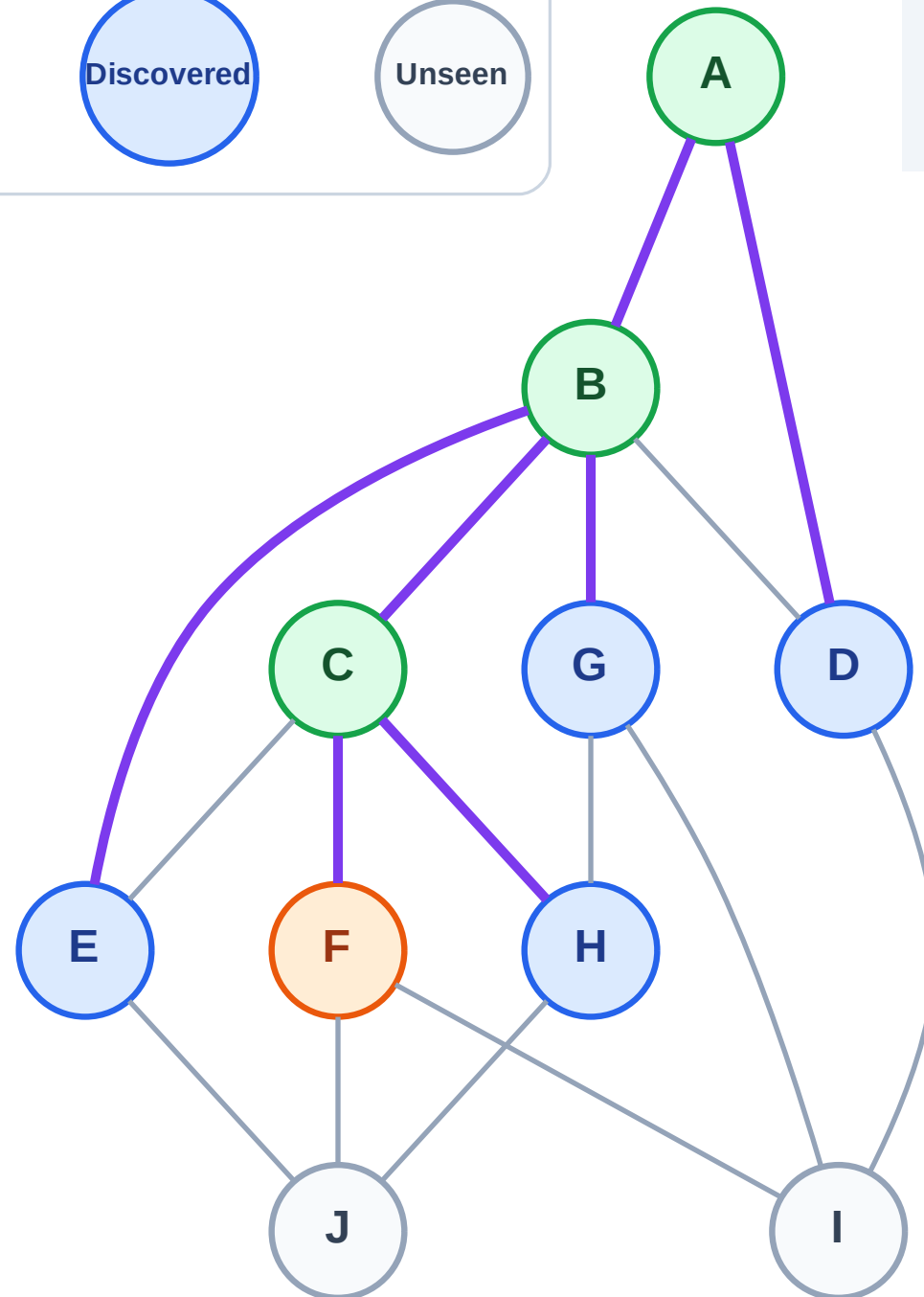
Step 8: Process node F

Step 8: Process node F



Stack (top at right): D | G | E | H

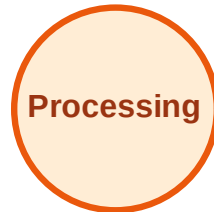
Visited: A, B, C



DFS Traversal

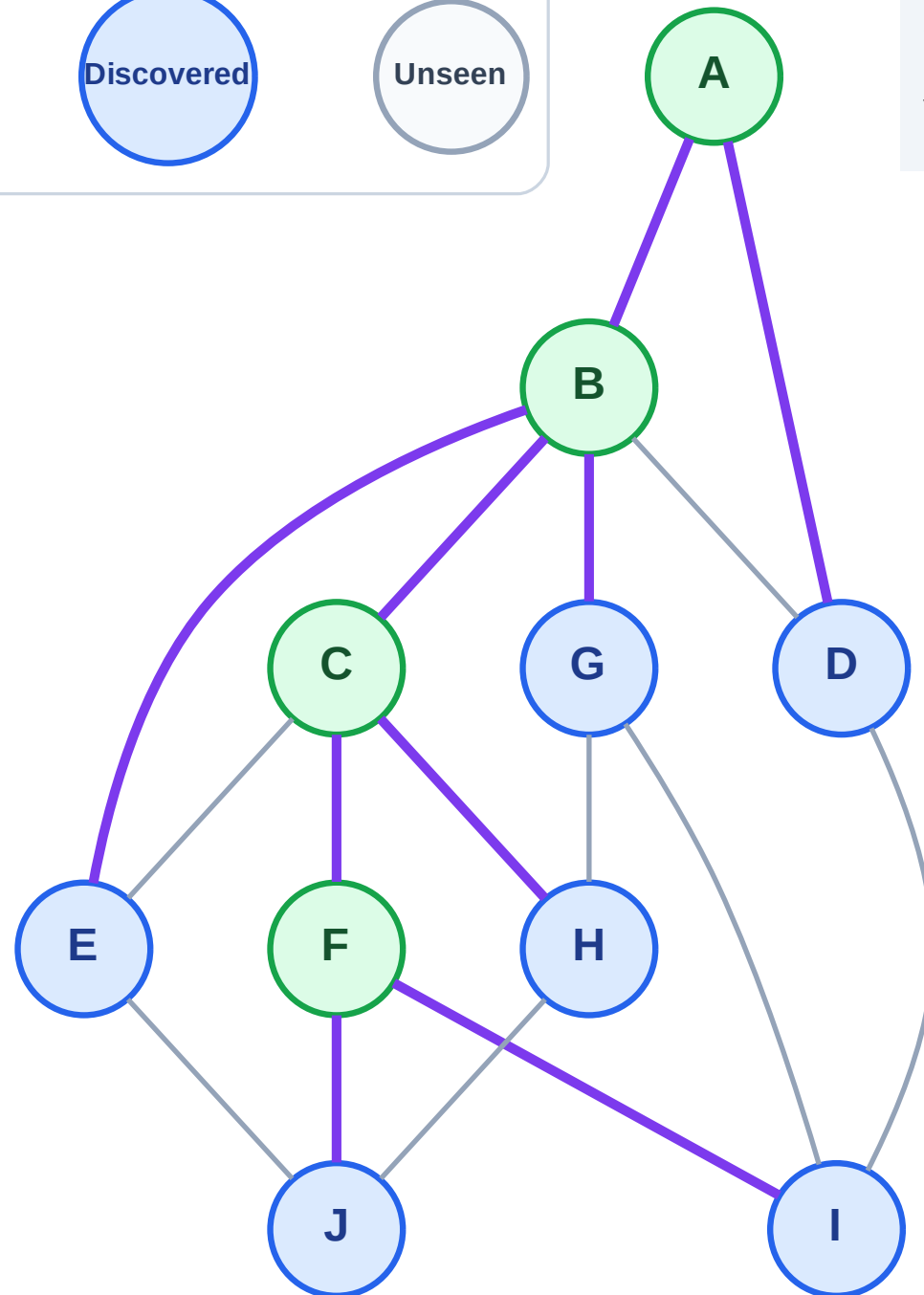
Step 9: Discover J, I from F

Legend



Stack (top at right): D | G | E | H | J | I

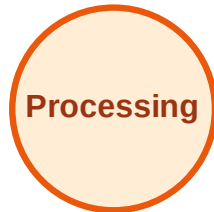
Visited: A, B, C, F



DFS Traversal

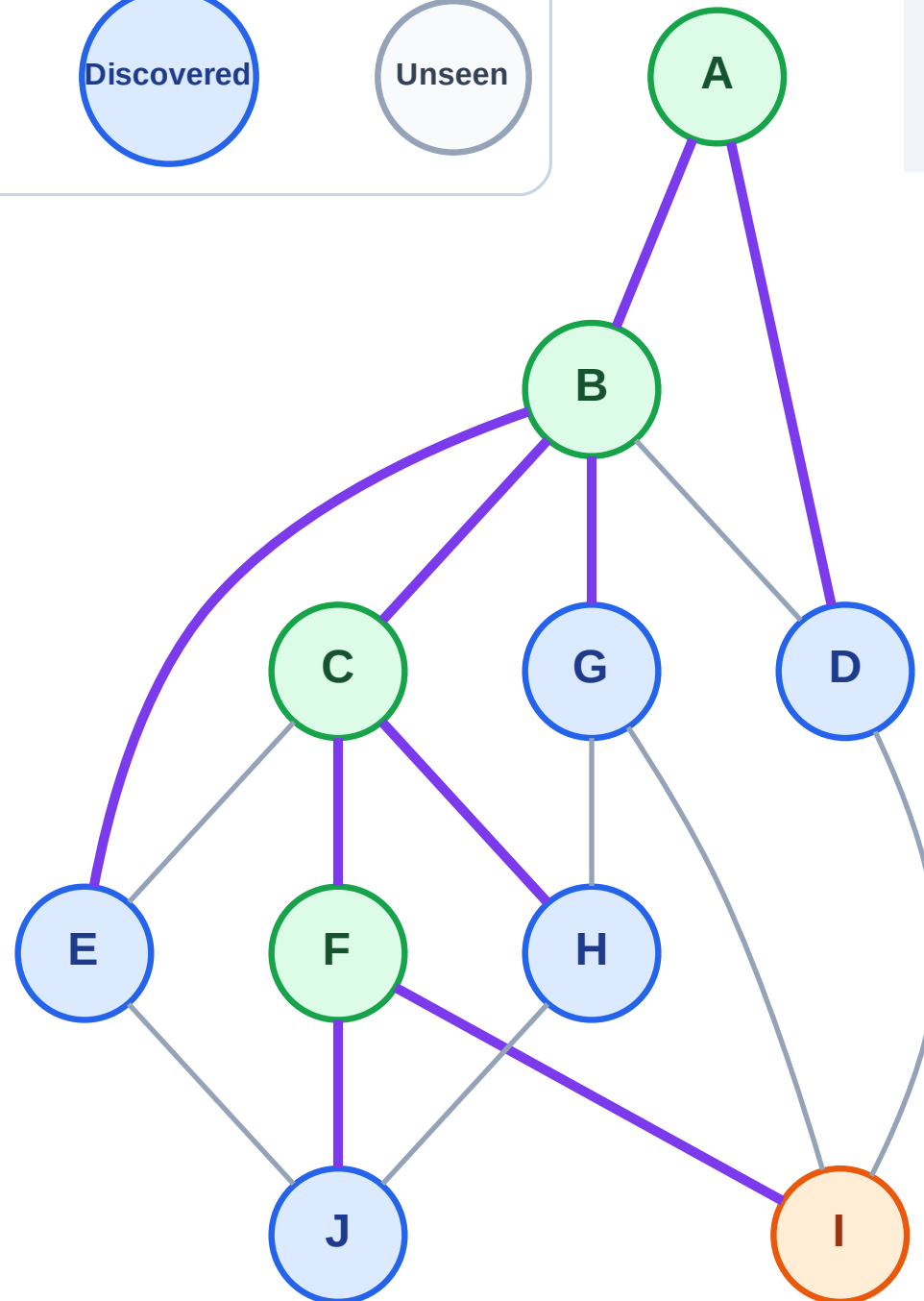
Step 10: Process node I

Legend



Stack (top at right): D | G | E | H | J

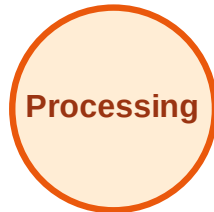
Visited: A, B, C, F



DFS Traversal

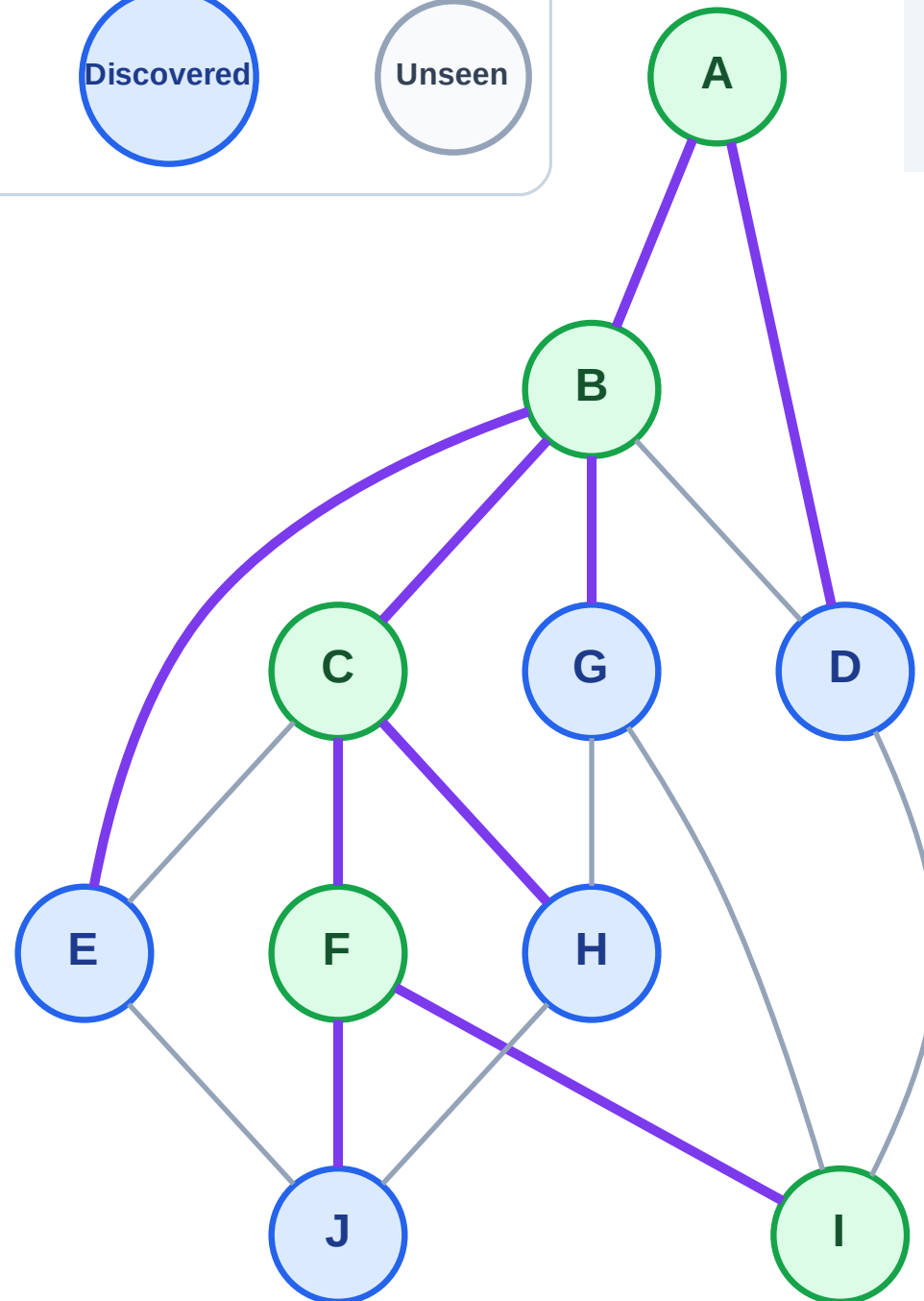
Step 11: No new neighbors from I

Legend



Stack (top at right): D | G | E | H | J

Visited: A, B, C, F, I



DFS Traversal

Step 12: Process node J

Legend

Complete

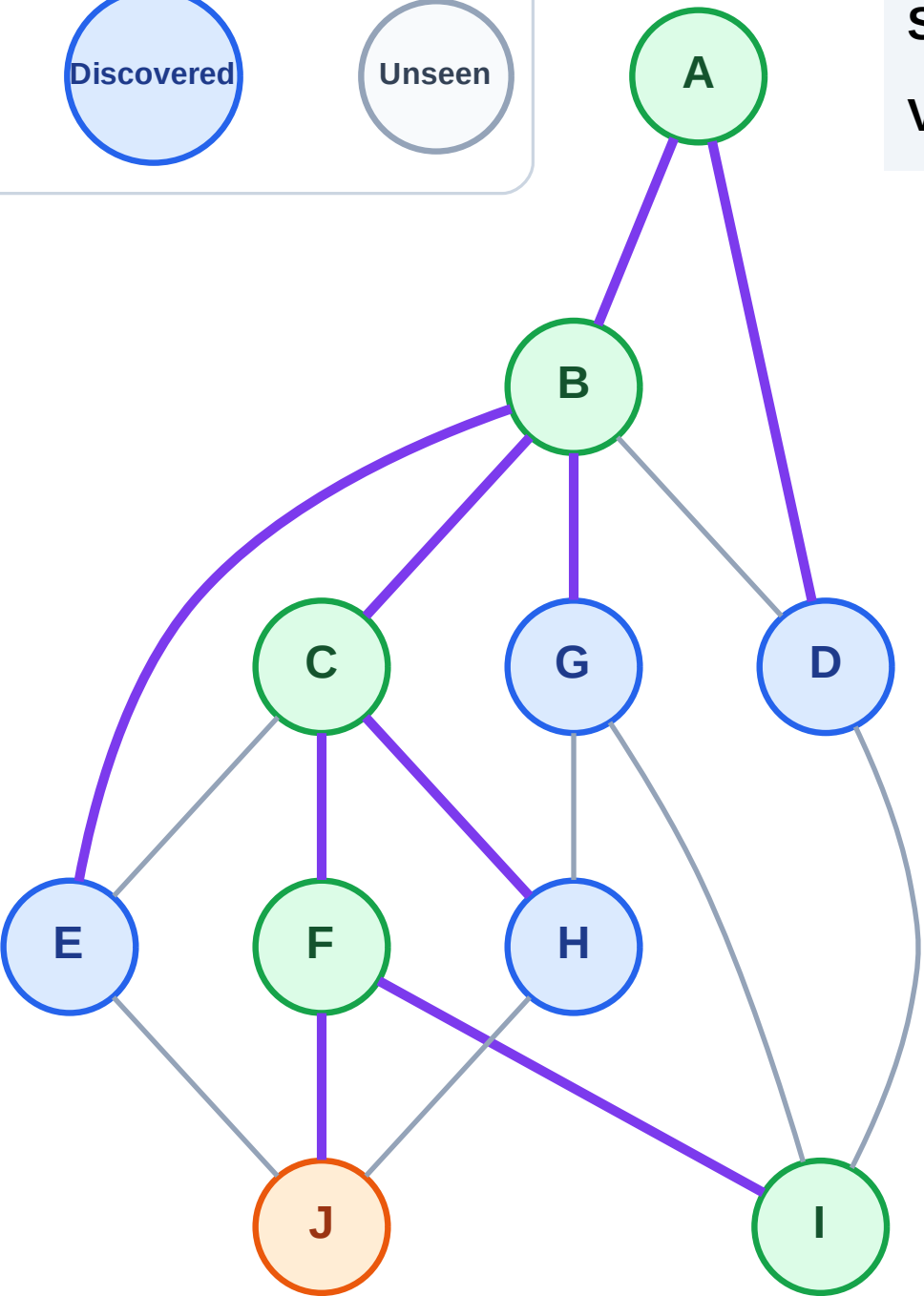
Processing

Discovered

Unseen

Stack (top at right): D | G | E | H

Visited: A, B, C, F, I



DFS Traversal

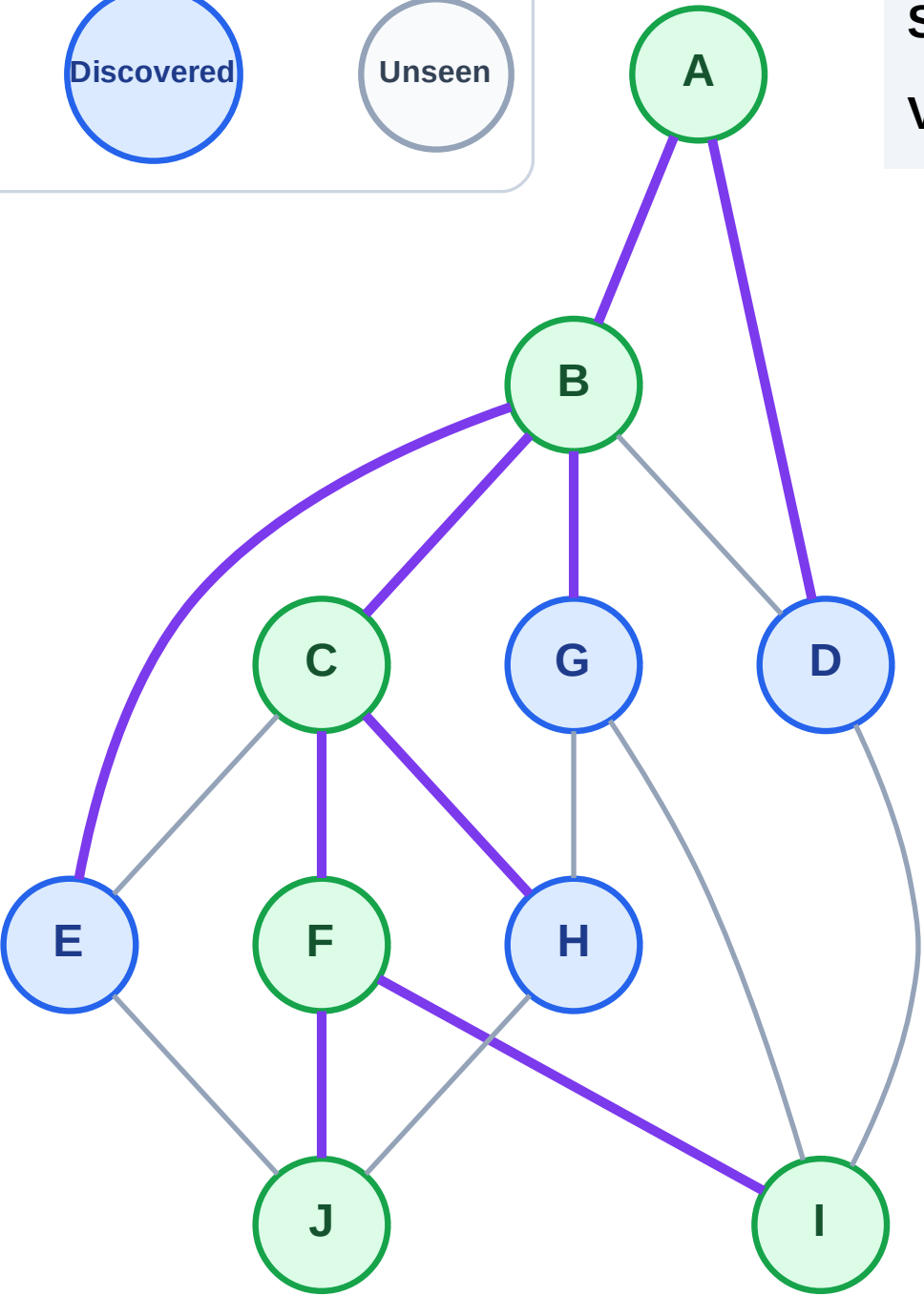
Step 13: No new neighbors from J

Legend



Stack (top at right): D | G | E | H

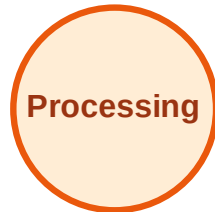
Visited: A, B, C, F, I, J



DFS Traversal

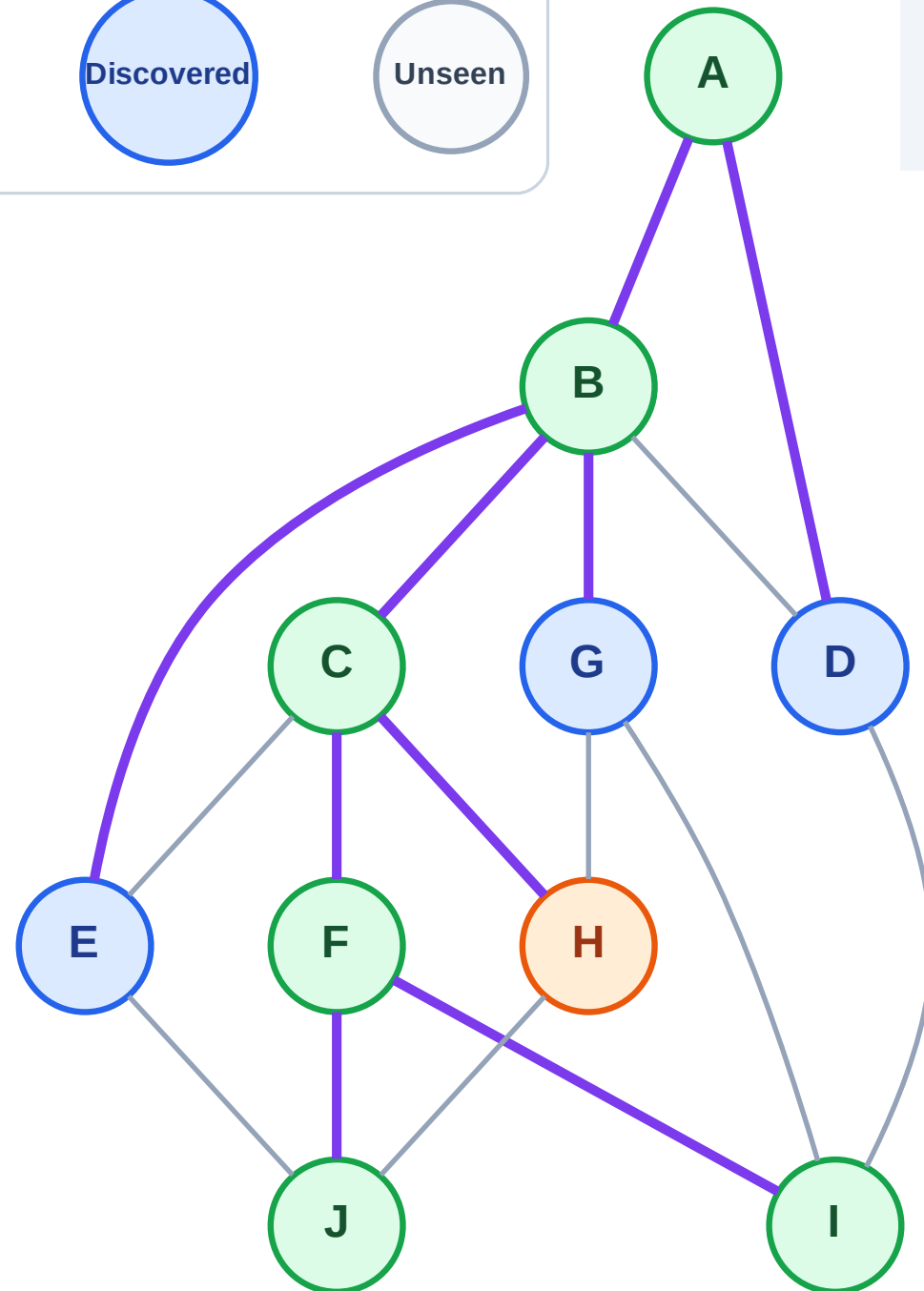
Step 14: Process node H

Legend



Stack (top at right): D | G | E

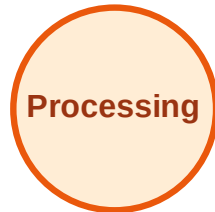
Visited: A, B, C, F, I, J



DFS Traversal

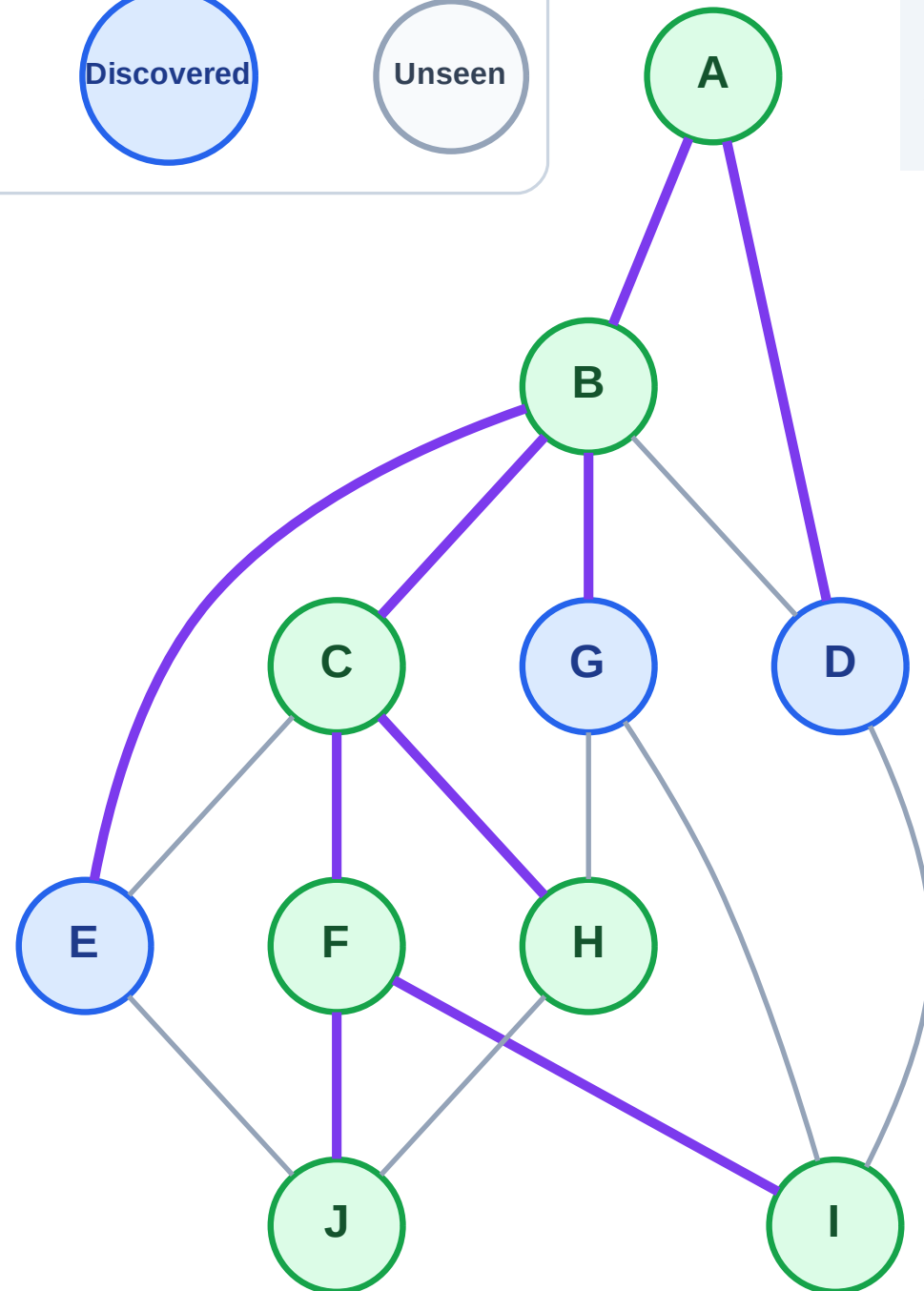
Step 15: No new neighbors from H

Legend



Stack (top at right): D | G | E

Visited: A, B, C, F, H, I, J



DFS Traversal

Step 16: Process node E

Legend

Complete

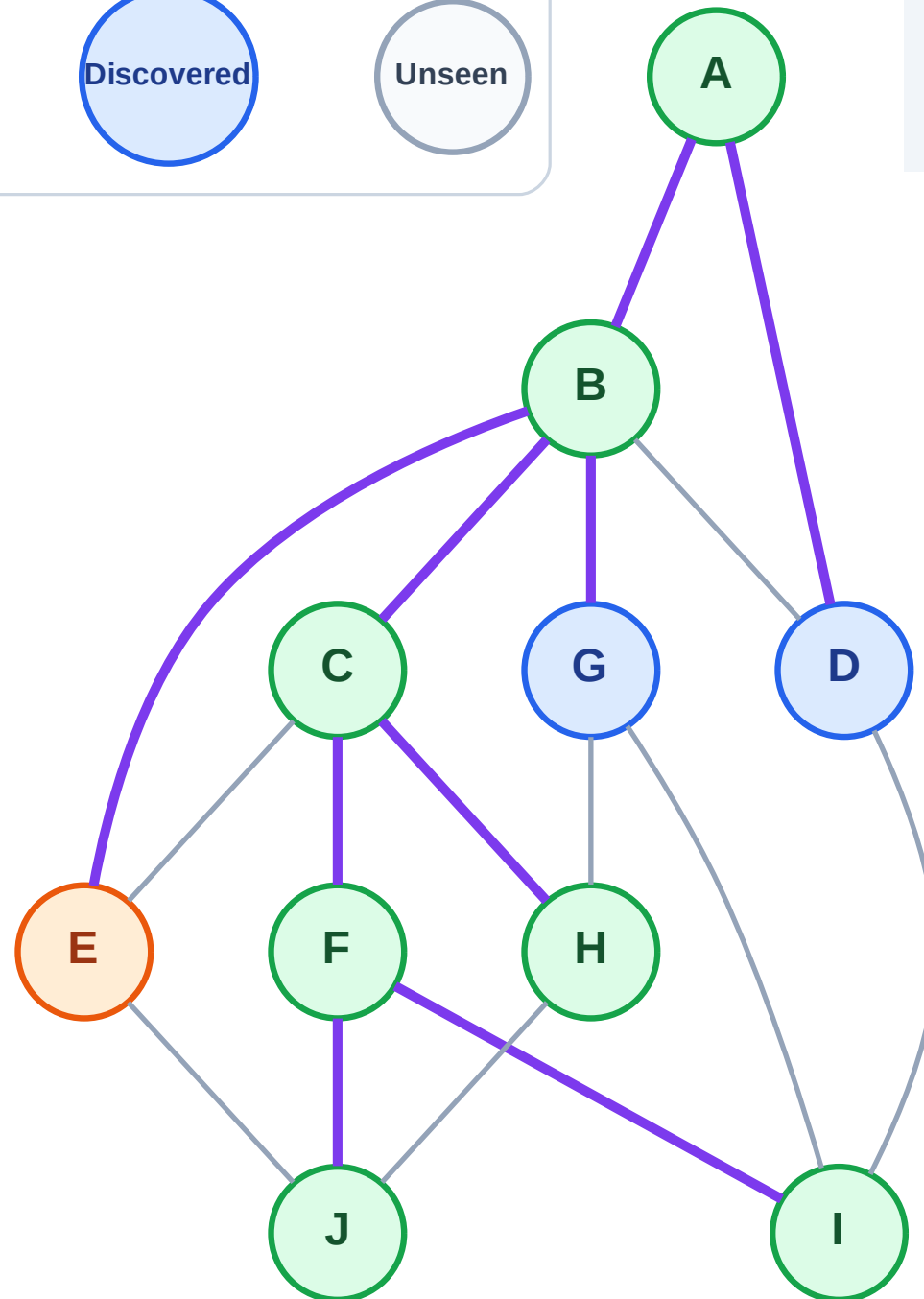
Processing

Discovered

Unseen

Stack (top at right): D | G

Visited: A, B, C, F, H, I, J



DFS Traversal

Step 17: No new neighbors from E

Legend

Complete

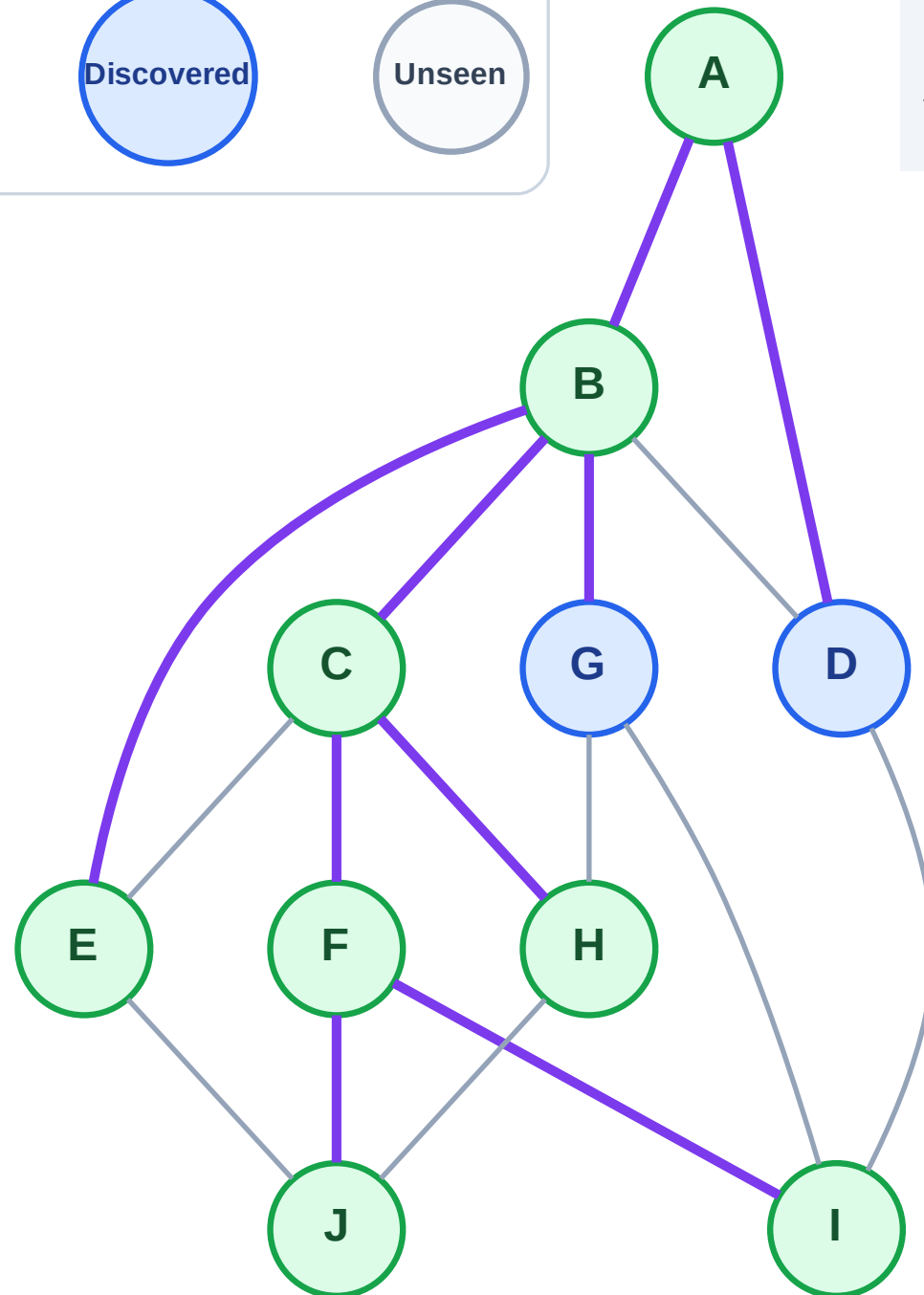
Processing

Discovered

Unseen

Stack (top at right): D | G

Visited: A, B, C, E, F, H, I, J



DFS Traversal

Step 18: Process node G

Legend

Complete

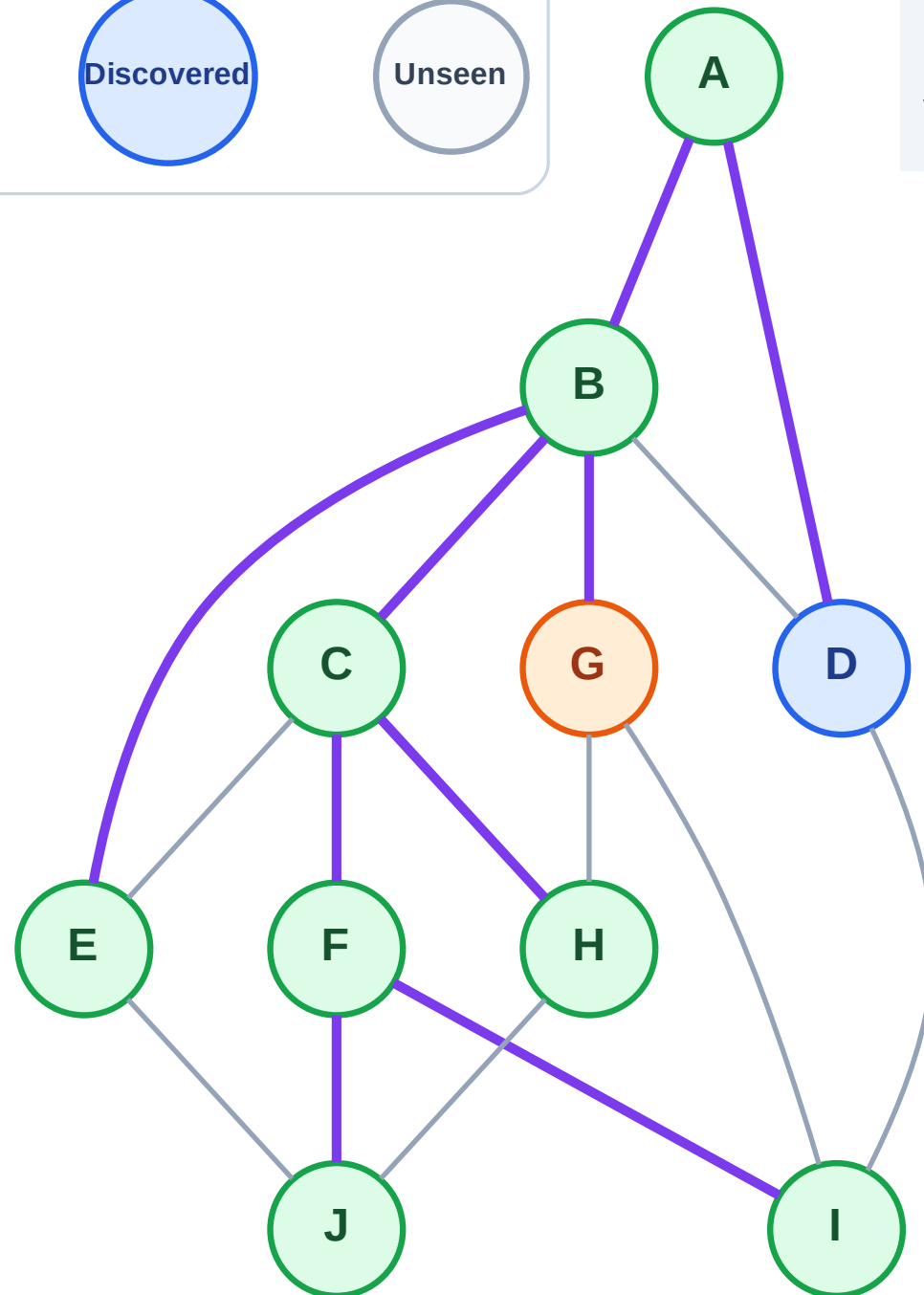
Processing

Discovered

Unseen

Stack (top at right): D

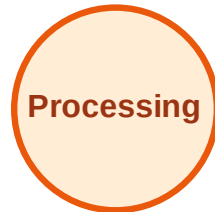
Visited: A, B, C, E, F, H, I, J



DFS Traversal

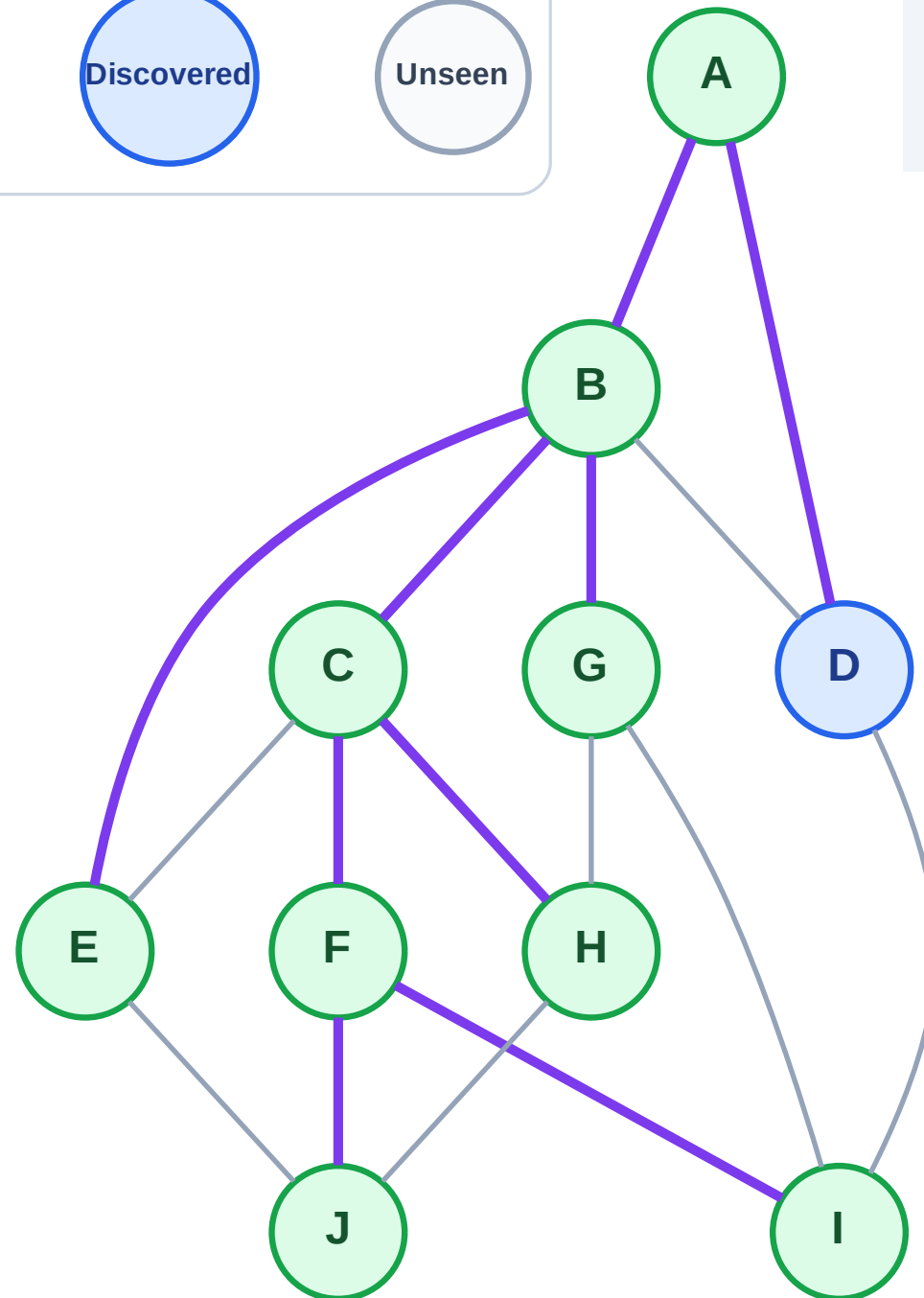
Step 19: No new neighbors from G

Legend



Stack (top at right): D

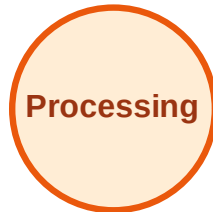
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

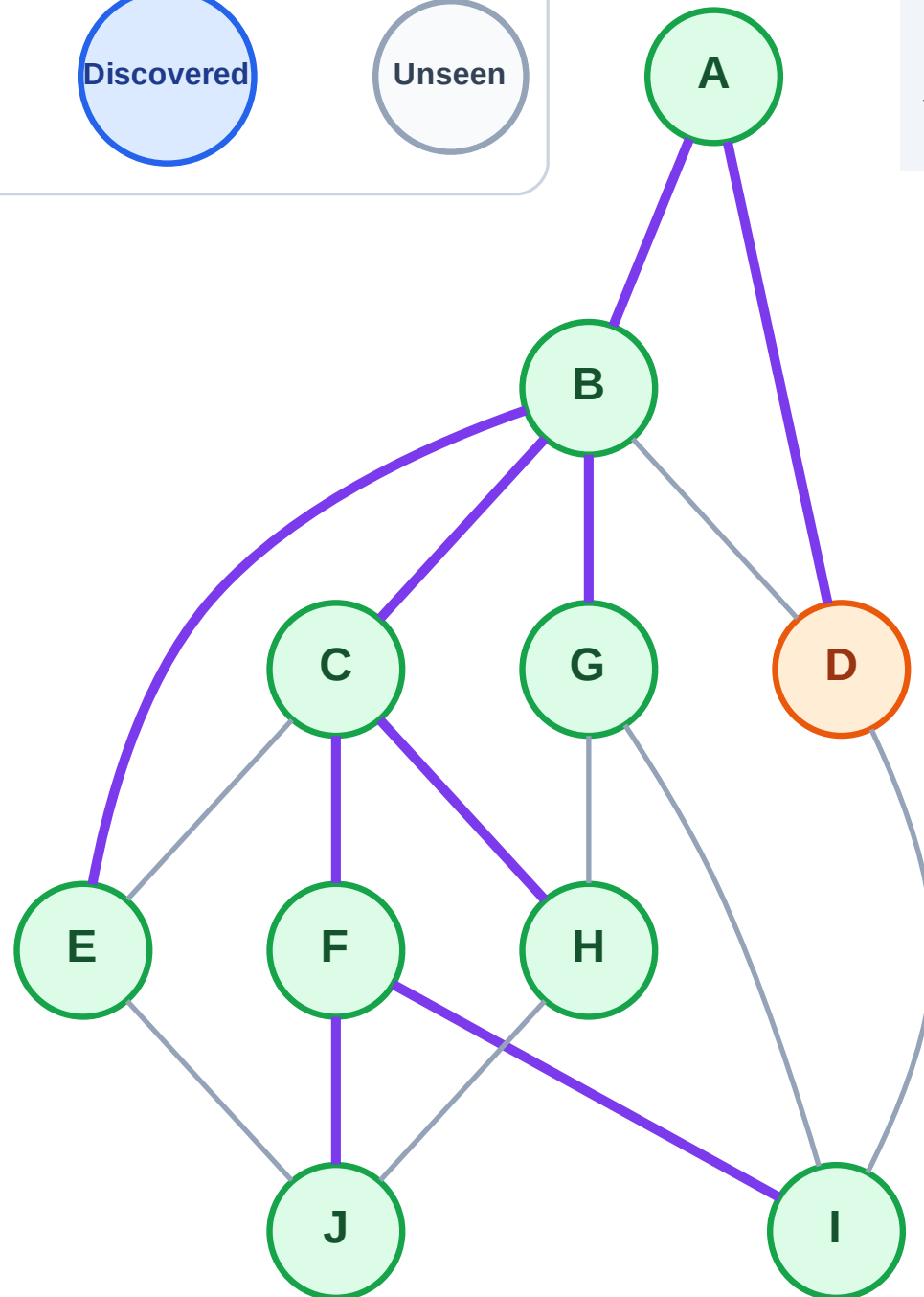
Step 20: Process node D

Legend



Stack (top at right): (empty)

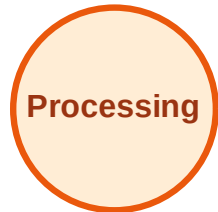
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from D

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

